

MILLER INDICES

Miller (& Miller-Bravais) Indices for

- ❑ Planes & Directions
- ❑ Lattices & Crystals

Part of

**MATERIALS SCIENCE
& A Learner's Guide
ENGINEERING**

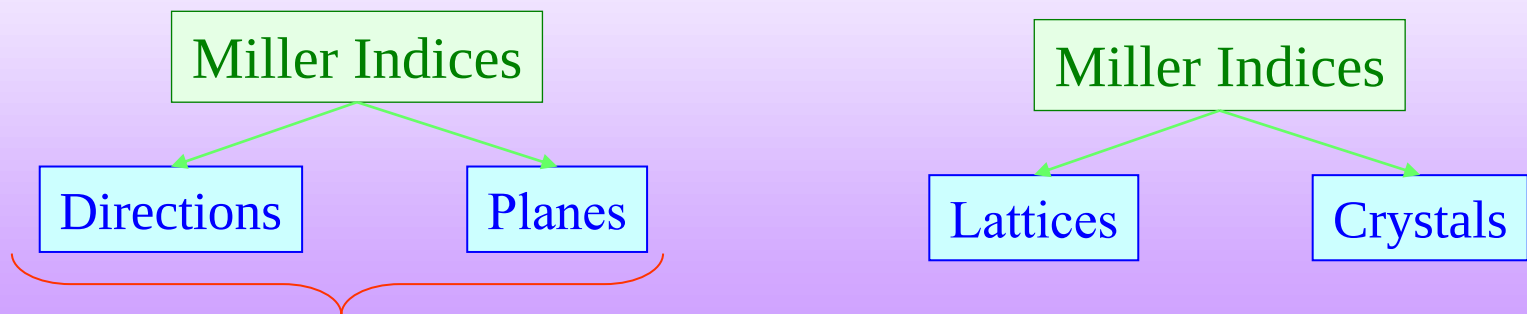
AN INTRODUCTORY E-BOOK

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<http://home.iitk.ac.in/~anandh/E-book.htm>

Note: in this book we have used '−' (minus) instead of a bar $\bar{}$ (i.e. $[-1\ 0\ 0]$ instead of $[\bar{1}\ 0\ 0]$) (this is technically wrong, but has been used for ease of typing)

From the law of rational indices developed by French Physicist and mineralogist
Abbé René Just Haüy
and popularized by
William Hallowes Miller

- ❑ Miller indices are used to specify **directions** and **planes**.
- ❑ These directions and planes could be in lattices or in crystals.
- ❑ *(It should be mentioned at the outset that special care should be given to see if the indices are in a lattice or a crystal).*
- ❑ The number of indices will match with the dimension of the lattice or the crystal: in **1D there will be 1 index**, in **2D there will be two indices**, in 3D there will be 3 indices, etc.
- ❑ Sometimes, like in the case of **Miller-Bravais indices** for **hexagonal** lattices and crystals, additional indices are used to highlight the symmetry of the structure. In the case of the Miller-Bravais indices for hexagonal structures, a third redundant index is added ($h\ k\ i\ l$)→ 4 indices are used in 3D space. The use of such redundant indices bring out the *equivalence* of the members of a ‘family’.
- ❑ Some aspects of Miller indices, especially those for planes, are not intuitively understood and hence some time has to be spent to familiarize oneself with the notation.



Note: both directions and planes are imaginary constructs

Miller indices for DIRECTIONS

- A vector $\mathbf{r}$ passing from the origin to a lattice point can be written as:

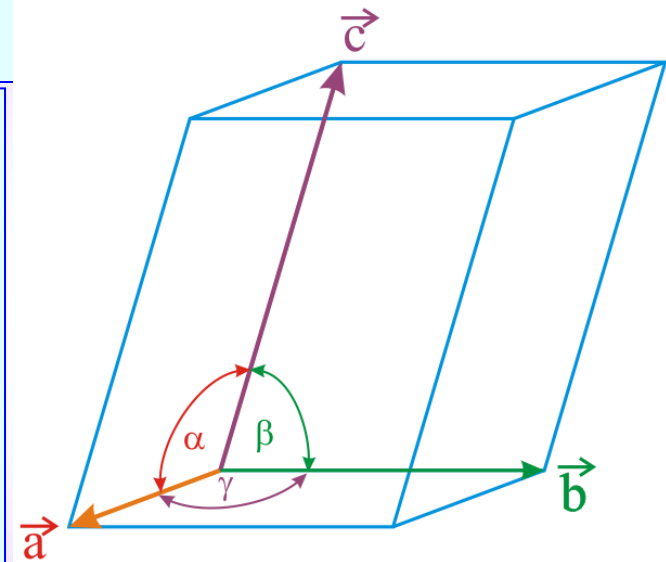
$$\mathbf{r} = r_1 \mathbf{a} + r_2 \mathbf{b} + r_3 \mathbf{c}$$

- Where, $\mathbf{a}, \mathbf{b}, \mathbf{c} \rightarrow$ basic vectors (or generator vectors).

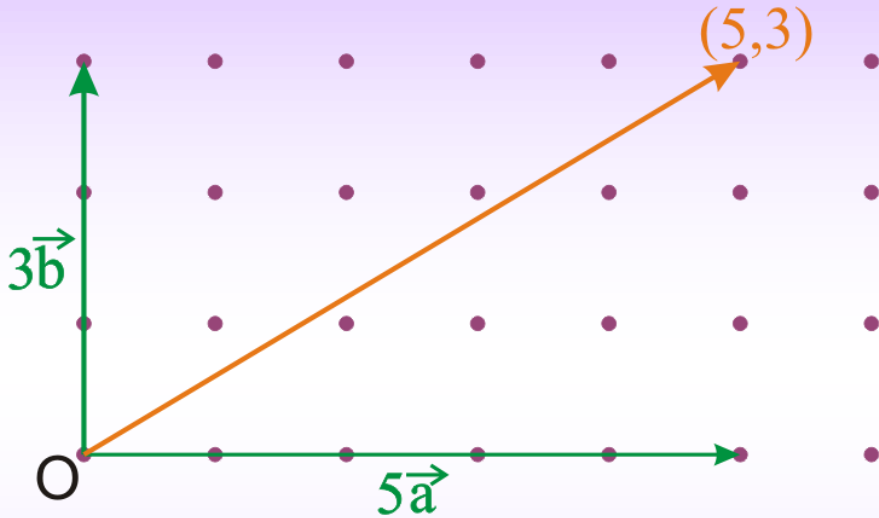
$$\mathbf{r} = r_1 \vec{a} + r_2 \vec{b} + r_3 \vec{c}$$

- Basis vectors are unit **lattice translation vectors**, which define the coordinate axis (as in the figure below).

- Note that their lengths are usually *one lattice translation* and **not 1 lengthscale unit!** (this is unlike for the basis vectors of a coordinate axis). To give an example, if a rectangle crystal has lattice parameters $a = 1$ cm and $b = 2.5$ cm, then $|\mathbf{a}| = 1$ cm and $|\mathbf{b}| = 2.5$ cm (it is not 1 cm along the axes and the scale of the unit along the two directions are different).
- In some cases, based on convenience, we may chose the basis vector as ‘multiple lattice translations’ (i.e. instead of one lattice translation we may chose 2 or 3).
- We may also chose alternate basis vectors for the same **structure** (which corresponds to different unit cells).

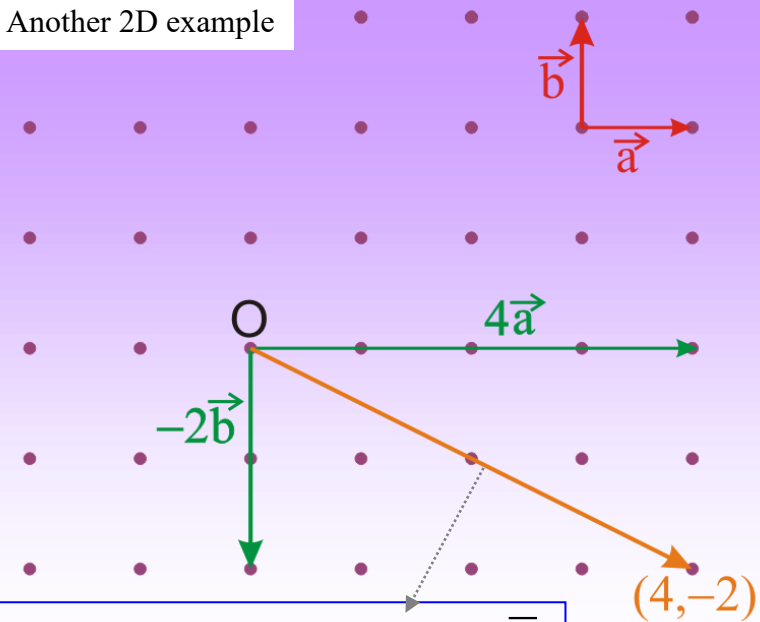


Miller Indices for directions in 2D



Miller indices $\rightarrow [53]$

Another 2D example



This direction is $\rightarrow [4\bar{2}]$

Normally, we 'take out' the common factor

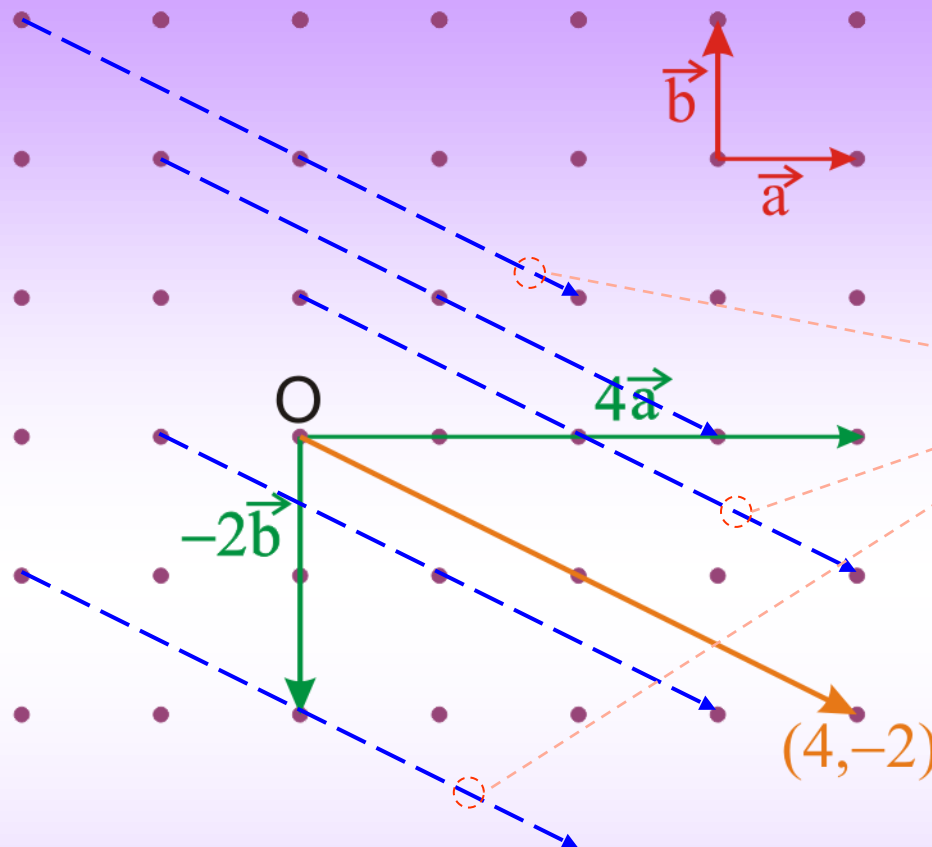
Miller Indices with magnitude $\rightarrow 2[2\bar{1}]$

And then omit it! Miller Indices $\rightarrow [2\bar{1}]$

STEPS in the determination of Miller indices for directions

- Position the vector, such that start (S: (x_1, y_1)) and end points (E: (x_2, y_2)) are lattice points and note the value of the coordinates. Subtract to obtain: $((x_2 - x_1), (y_2 - y_1))$.
- Write these number in square brackets, without the 'comma': $[* *]$.
- 'Remove' the common factors. (Note: keep the common factor, preferably outside the bracket, if the length has to be preserved in further computations).

We will see an example soon



Set of directions represented
by the Miller index

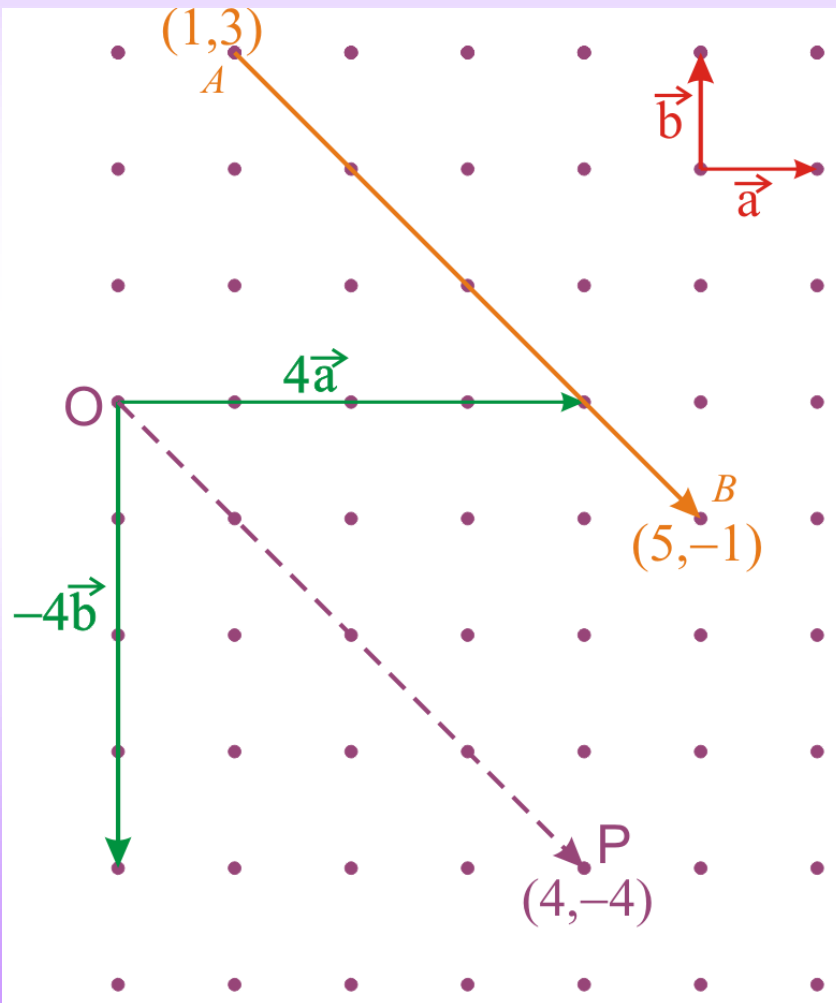
$$[4\bar{2}] = 2[2\bar{1}]$$

Miller Indices $\rightarrow 2[2\bar{1}]$

This **Miller index** represents a set (an infinite set) of all such parallel vectors (*and not just one vector*)
(Note: 'usually' (actually always for now!))
originating at a lattice point and ending at a lattice point

How to find the Miller Indices for an arbitrary direction? → Procedure

- ❑ Consider the example below.
- ❑ Subtract the coordinates of the end point from the starting point of the vector denoting the direction → If the starting point is A(1,3) and the final point is B(5,-1) → the difference (B-A) would be (4, -4).



- ❑ Enclose in square brackets, remove comma and write negative numbers with a bar → $[4\bar{4}]$
- ❑ Factor out the common factor → $4[1\bar{1}]$
- ❑ If we are worried about the direction and magnitude then we write → $4[1\bar{1}]$
- ❑ If we consider only the direction then we write → $[1\bar{1}]$
- ❑ Needless to say the first vector is 4 times in length
- ❑ The magnitude of the vector $[1\bar{1}] = |[1\bar{1}]|$ is $\sqrt{(1)^2 + (-1)^2} = \sqrt{2}$

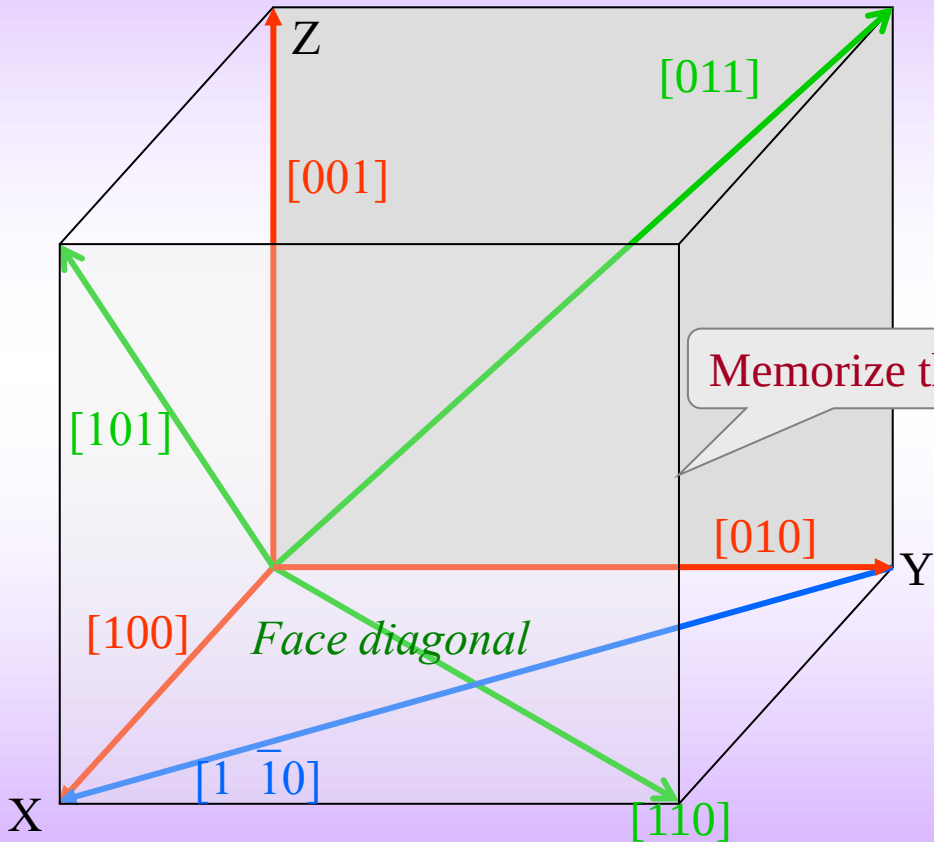
Further points

❑ General Miller indices for a direction in 3D is written as $[u\ v\ w]$.

❑ The length of the vector represented by the Miller indices is:

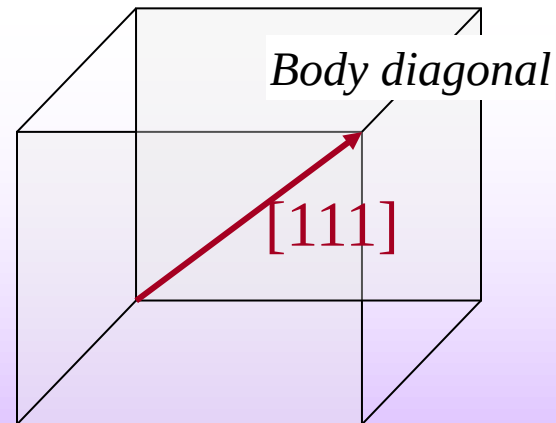
$$\sqrt{u^2 + v^2 + w^2}$$

Important directions in 3D represented by Miller Indices (cubic lattice)



■ Three important set of directions related to the cube are:

- (i) cell edge ($[001]$ type),
- (ii) face diagonal ($[011]$ type),
- (iii) body diagonal ($[111]$ type).



Procedure as before→

- (Coordinates of the final point – coordinates of the initial point).
- Reduce to smallest integer values.

The concept of a family of directions

- ❑ A set of directions related by **symmetry operations** of the lattice or the crystal is called a family of directions. *A family is a symmetry related set.*
- ❑ A family of directions is represented (Miller Index notation) as: $\langle u \ v \ w \rangle$. *Note the brackets.*
- ❑ Hence one has to ask two questions before deciding on the list of the members of a family:
 - 1 ➤ *Is one considering the lattice or the crystal?*
 - 2 ➤ *What is the crystal system one is talking about. (What is its point group symmetry?)*

Miller indices for a direction in a lattice versus a crystal

- ❑ We have seen in the chapter on geometry of crystals that crystal can have symmetry equal to or lower than that of the lattice.
- ❑ If the symmetry of the crystal is lower than that of the lattice then two members belonging to the same family in a lattice **need not** belong to the same family in a crystal → this is because *crystals can have lower symmetry than a lattice* (examples which will taken up soon will explain this point).

*Let us consider some examples to understand the concept of family of directions.
You may want to revise chapter on [Symmetry](#) and [Making Crystals](#)'.*

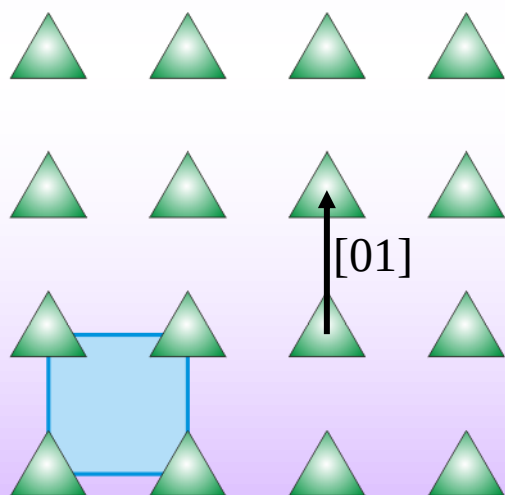
Let us start with a question.

Q & A

For the crystal in the figure below, what are the member(s) of the family of directions $\langle 01 \rangle$?

- ☐ The family $\langle 01 \rangle$ has only one member $\rightarrow [01]$!!!
- ☐ This crystal has m symmetry and belongs to the rectangle 2D crystal system. (*Not a square crystal, though the unit cell is a square*).
- ☐ The mirrors are vertical and pass through lattice points and between them.

[Click here to know more](#)



Family of directions

Examples

Let us consider a **square lattice**:

- $[10]$ and $[01]$ belong to the same family $\rightarrow$ related by a 4-fold rotation
- $[11]$ and $[\bar{1}\bar{1}]$ belong to the same family $\rightarrow$ related by a 4-fold rotation
- $[01]$ and $[0\bar{1}]$ belong to the same family $\rightarrow$ related by a 2-fold rotation

(or double action of 4-fold)

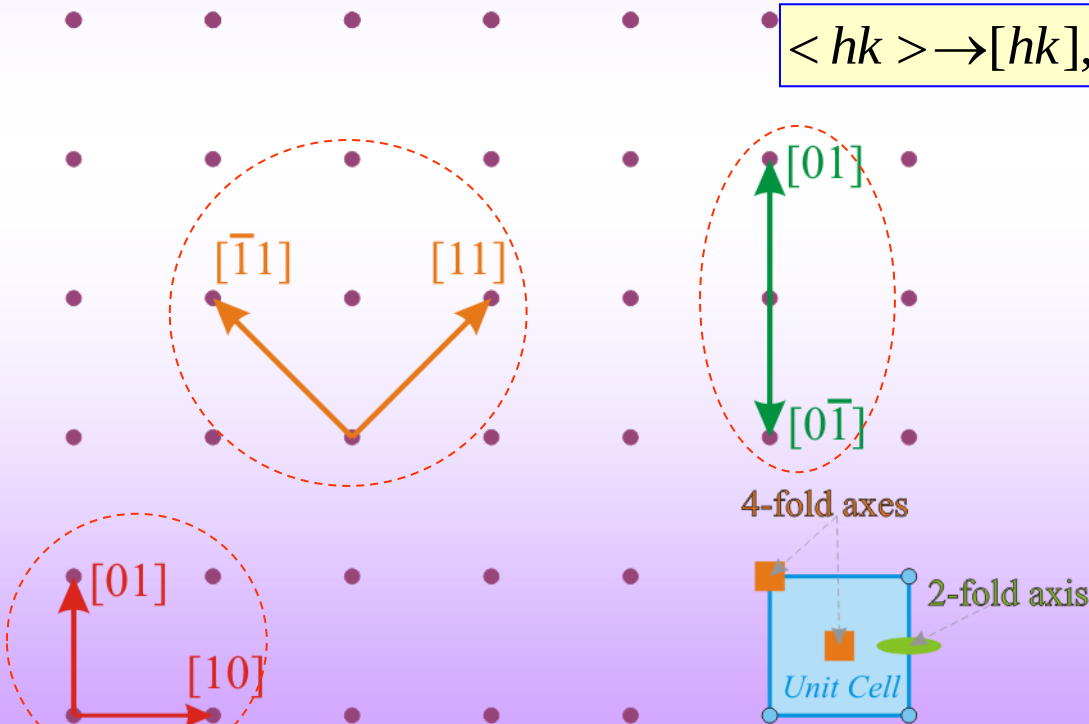
Writing down all the members of the family

$$\langle hk \rangle \rightarrow [hk], [\bar{h}k], [h\bar{k}], [\bar{h}\bar{k}], [kh], [k\bar{h}], [\bar{k}h], [\bar{k}\bar{h}]$$

$$\langle 10 \rangle \rightarrow [10], [01], [\bar{1}0], [0\bar{1}]$$

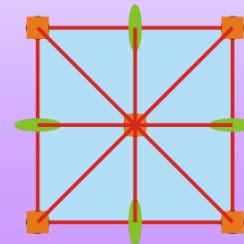
$$\langle 11 \rangle \rightarrow [11], [\bar{1}\bar{1}], [1\bar{1}], [\bar{1}1]$$

Essentially the 1st and 2nd index can be interchanged and be made negative (due to high symmetry)



For sake of clarity all symmetry operators have not been marked

4mm



Let us consider a **Rectangle** lattice:

- $[10]$ and $[01]$ do **NOT** belong to the same family
- $[11]$ and $[\bar{1}1]$ belong to the same family $\rightarrow$ related by a mirror
- $[01]$ and $[0\bar{1}]$ belong to the same family $\rightarrow$ related by a 2-fold rotation
- $[21]$ and $[12]$ do **NOT** belong to the same family

Writing down all the members of the family

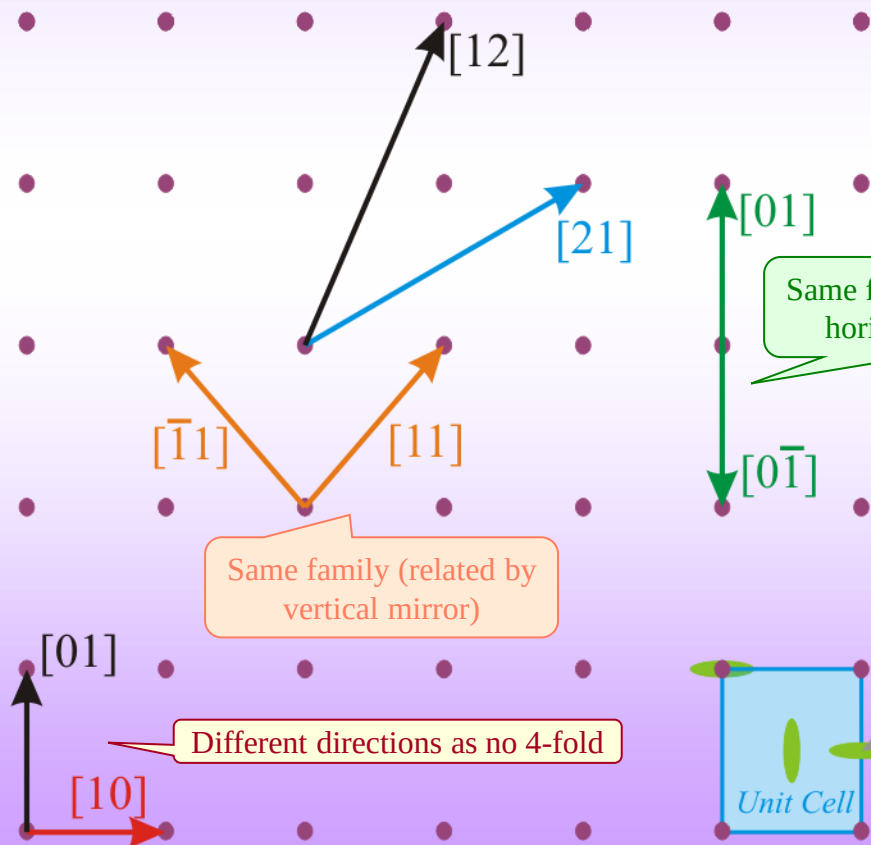
$$\langle hk \rangle \rightarrow [hk], [\bar{h}k], [h\bar{k}], [\bar{h}\bar{k}]$$

$$\langle 10 \rangle \rightarrow [10], [\bar{1}0]$$

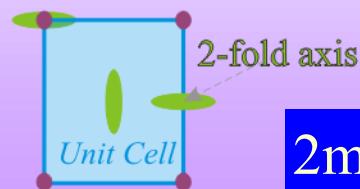
$$\langle 11 \rangle \rightarrow [11], [\bar{1}1], [1\bar{1}], [\bar{1}\bar{1}]$$

$$\langle 12 \rangle \rightarrow [12], [\bar{1}2], [1\bar{2}], [\bar{1}\bar{2}]$$

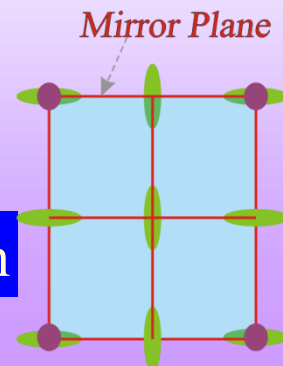
The 1st and 2nd index can **NOT** be interchanged, but can be made negative



For sake of clarity all symmetry axes have not been marked



2mm



Let us consider a square lattice decorated with a rotated square to give a **SQUARE CRYSTAL** (*as 4-fold still present*):

- 

Writing down all the members of the family

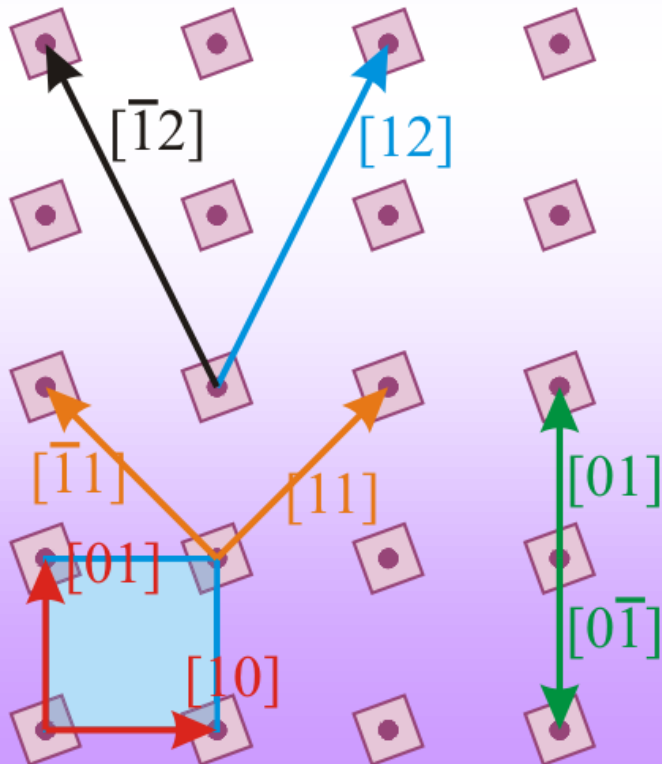
$$\langle hk \rangle \rightarrow [hk], [\bar{h} \bar{k}], [\bar{k}h], [k\bar{h}]$$

$$\langle 10 \rangle \rightarrow [10], [\bar{1}0], [01], [0\bar{1}]$$

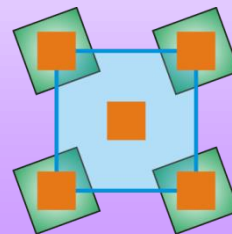
$$\langle 11 \rangle \rightarrow [11], [\bar{1}1], [1\bar{1}], [\bar{1}\bar{1}]$$

$$\langle 12 \rangle \rightarrow [12], [\bar{2}1], [\bar{1}\bar{2}], [2\bar{1}]$$

$$\langle 21 \rangle \rightarrow [21], [\bar{1}2], [\bar{2}\bar{1}], [1\bar{2}]$$



4

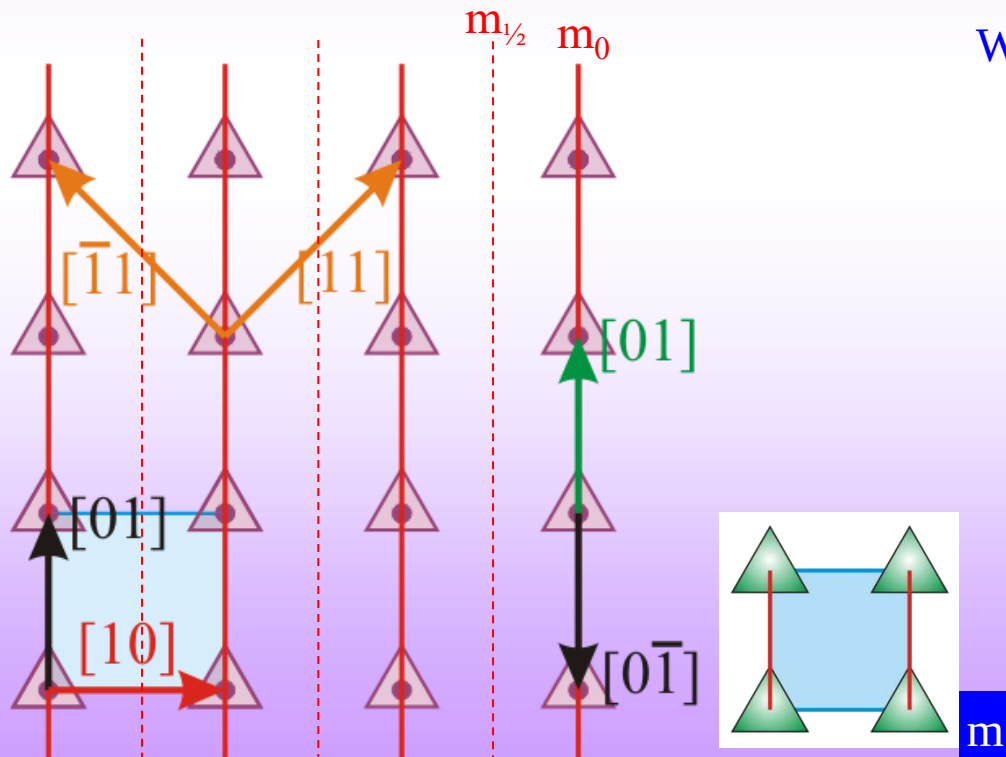


No Mirrors

Let us consider a square lattice decorated with a triangle to give a **RECTANGLE CRYSTAL**:

- $[10]$ and $[01]$ do **NOT** belong to the same family
→ 4-fold rotation destroyed in the crystal
- $[11]$ and $[\bar{1}\bar{1}]$ belong to the same family → related by mirror
- $[11]$ and $[1\bar{1}]$ do **NOT** belong to the same family
- $[01]$ and $[0\bar{1}]$ do **NOT** belong to the same family

Thought
provoking
example



Writing down all the members of the family

$$\langle hk \rangle \rightarrow [hk], [\bar{h}\bar{k}]$$

$$\langle 10 \rangle \rightarrow [10], [\bar{1}0]$$

$$\langle 01 \rangle \rightarrow [01]$$

$$\langle 0\bar{1} \rangle \rightarrow [0\bar{1}]$$

$$\langle 11 \rangle \rightarrow [11], [\bar{1}\bar{1}]$$

$$\langle 1\bar{1} \rangle \rightarrow [1\bar{1}], [\bar{1}1]$$

Important Note

Hence, all directions related by symmetry (only) form a family

Family of directions

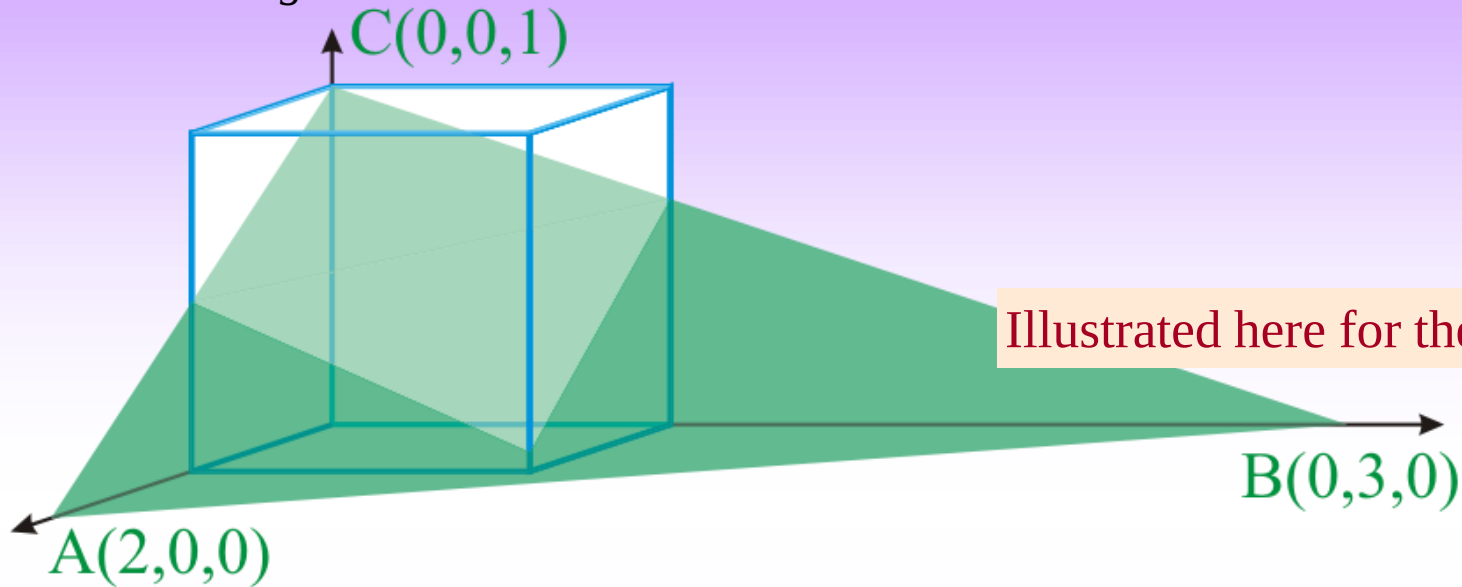
Index	Members in family for cubic lattices <i>(True for cubic crystals in holohedral class, but not true for all cubic crystals)</i>	Number
<100>	[100], [$\bar{1}$ 00], [010], [0 $\bar{1}$ 0], [001], [00 $\bar{1}$]	$3 \times 2 = 6$
<110>	[110], [$\bar{1}$ 10], [1 $\bar{1}$ 0], [$\bar{1}$ $\bar{1}$ 0], [101], [$\bar{1}$ 01], [10 $\bar{1}$], [$\bar{1}$ 0 $\bar{1}$], [011], [0 $\bar{1}$ 1], [01 $\bar{1}$], [0 $\bar{1}$ $\bar{1}$]	$6 \times 2 = 12$
<111>	[111], [$\bar{1}$ 11], [1 $\bar{1}$ 1], [11 $\bar{1}$], [$\bar{1}$ $\bar{1}$ 1], [$\bar{1}$ 1 $\bar{1}$], [1 $\bar{1}$ $\bar{1}$], [$\bar{1}$ $\bar{1}$ $\bar{1}$]	$4 \times 2 = 8$

the 'negatives' (opposite direction)

Symbol	Alternate symbol		
[]		→	Particular direction
< >	[[]]	→	Family of directions

Miller Indices for PLANES

Miller indices for planes is not as intuitive as that for directions and special care must be taken in understanding them



- ❑ Find intercepts along axes $\rightarrow 2 \ 3 \ 1$
- ❑ Take reciprocal $\rightarrow \frac{1}{2} \ \frac{1}{3} \ 1^*$
- ❑ Convert to smallest integers in the same ratio $\rightarrow 3 \ 2 \ 6$
- ❑ Enclose in parenthesis $\rightarrow (326)$ *Note the type of brackets*
- Note: (326) does **NOT** represent one plane but an infinite **set** of parallel planes passing through lattice points.
- Set of planes should **not** be confused with a **family** of planes- which we shall consider next.

* As we shall see later – reciprocals are taken to avoid infinities in the 'defining indices' of planes



- Why do we need Miller indices (say for planes)?
- Can't we just use intercepts to designate planes?

Thus we see that Miller indices does the following:

- ❑ *Avoids infinities in the indices* (intercepts of $(1, \infty, \infty)$ becomes (100) index).
- ❑ Avoids dimensioned numbers
 - Instead we have multiples of lattice parameters along the **a**, **b**, **c** directions (this implies that $1a$ could be $10.2\text{\AA}$, while $2b$ could be $8.2\text{\AA}$).

- How to know if a given atom/lattice point is 'sitting' on a plane?

- ❑ Use the intercept form of the equation of a plane. If $(a,0,0)$, $(0,b,0)$ & $(0,0,c)$ are the intercepts along x, y & z, then the equation is as below.
- ❑ Substitute the coordinates of the atom to check if equation is satisfied [done for the $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ atom of BCC as below (for the (112) plane).

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \text{The } (112) \text{ plane} \rightarrow x + y + 2z = 1$$

$$\text{Check for the } (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \text{ atom} \rightarrow (\frac{1}{2} + \frac{1}{2} + 2 \cdot \frac{1}{2} - 1) \neq 0 \quad \text{Check} \rightarrow (\frac{1}{2} + \frac{1}{2} - 1) = 0$$

This implies that the atom at $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ does not sit on this plane.

This $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ atom sits on the (110) plane

$$\text{The } (110) \text{ plane} \rightarrow x + y = 1$$

$$\text{Check} \rightarrow (\frac{1}{2} + \frac{1}{2} - 1) = 0$$

Note: as done previously, we will continue to call planes in lower dimensions (like 2D) as planes—though they are actually lines in 2D.

The concept of a family of planes

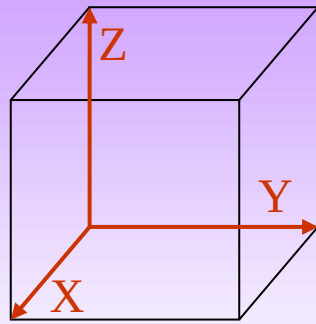
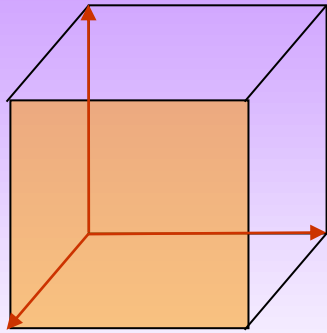
- ❑ A set of planes related by **symmetry operations of the lattice or the crystal** is called a family of planes (the translation symmetry operator is excluded→ the translational symmetry is included in the definition of a plane itself*).
- ❑ All the points which one should keep in mind while dealing with directions to get the members of a family, should also be kept in mind when dealing with planes.
- ❑ **Members of a family (of planes or directions) of a lattice or crystal, as they are related by the symmetry of the crystal, are identical (in all respects).**
- ❑ The family of planes is enclosed in **{ }** brackets, while an individual member is enclosed in () brackets.

** As the Miller index for a plane line (100) implies a infinite parallel set of planes.*

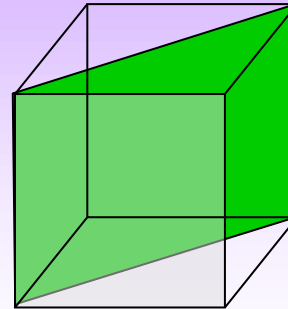
Cubic lattice

Let us consider a cubic lattice and important (low index) planes therein).

These planes are $\{100\}$, $\{110\}$ & $\{111\}$.



Note: Do NOT pass plane through origin. Shift it by one unit in x,y,z or a combination.



Intercepts $\rightarrow 1 \ 1 \ \infty$
 Plane $\rightarrow (110)$ (reciprocal $\rightarrow 1 \ 1 \ 0$)
 Family $\rightarrow \{110\} \rightarrow 12 \text{ members}$

For the orange plane

Intercepts $\rightarrow 1 \ \infty \ \infty$ (reciprocal $\rightarrow 1 \ 0 \ 0$)

Plane $\rightarrow (100)$

Family $\rightarrow \{100\} \rightarrow 6 \text{ members}$

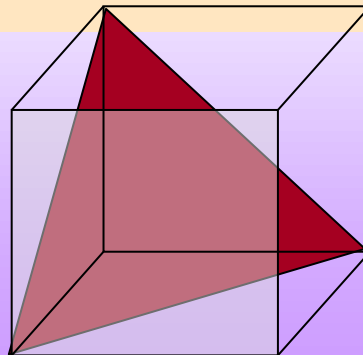
Members: $(100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$

Obtained by using the point group symmetry: $4/m \bar{3} 2/m$

Members: $(110), (1\bar{1}0), (011), (0\bar{1}1), (\bar{1}10), (\bar{1}\bar{1}0), (101), (1\bar{0}1), (01\bar{1}), (0\bar{1}\bar{1}), (\bar{1}0\bar{1}), (\bar{1}\bar{0}\bar{1})$

Note: (100) and $(\bar{1}00)$ are identical in terms of Miller indices, as the common factor -1 can be 'removed'. Also, the infinite set of (100) planes includes the $(\bar{1}00)$ plane. However, often we retain both these while writing out the members of the family.

The purpose of using reciprocal of intercepts and not intercepts themselves in Miller indices becomes clear $\rightarrow$ the ∞ are removed



Intercepts $\rightarrow 1 \ 1 \ 1$
 Plane $\rightarrow (111)$ (reciprocal $\rightarrow 1 \ 1 \ 1$)
 Family $\rightarrow \{111\} \rightarrow 8$
 (Octahedral plane)

Members: $(111), (\bar{1}\bar{1}\bar{1}), (11\bar{1}), (1\bar{1}1), (\bar{1}11), (\bar{1}1\bar{1}), (1\bar{1}\bar{1}), (\bar{1}\bar{1}1)$

Points about planes and directions

- ❑ Typical representation of an unknown/general direction $\rightarrow [uvw]$.
Corresponding family of directions $\rightarrow \langle uvw \rangle$.
- ❑ Unknown/general plane $\rightarrow (hkl)$.
Corresponding family of planes $\rightarrow \{hkl\}$.
- ❑ Double digit indices should be separated by commas or spaces
 $\rightarrow (12,22,3)$ or $(12\ 22\ 3)$.
- ❑ In cubic lattices/crystals the (hkl) plane is perpendicular to the $[hkl]$ direction
(specific plane perpendicular to a specific direction)
➤ $[hkl] \perp (hkl)$. E.g. $[1-11] \perp (1-11)$.
However, this cannot be generalized to all crystals/lattices (*though there are other specific examples*).

Note: the correct way is to put a 'bar'.

- ❑ The inter-planar spacing can be computed knowing the Miller indices for the planes (hkl) and the kind of lattice involved. The formula to be used depends on the kind of lattice and the formula for cubic lattices is given below.

Interplanar spacing ($d_{(hkl)}$) in cubic lattice (& crystals)

$$d_{(hkl)}^{\text{cubic lattice}} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

❑ What does the ‘symbol’ (111) mean/represent?

The symbol (111) represents Miller indices for an *infinite set of parallel planes*, with intercepts 1, 1 & 1 along the three crystallographic axis (unit lattice parameter along these), *which pass through lattice points*.

❑ (111) is the Miller indices for a plane (?) (to reiterate)

➤ It is usually for an infinite *set* of parallel planes, with a specific ‘d’ spacing. Hence, (100) plane is no different from a (−100) plane (i.e. a set consists of planes related by translational symmetry).

However, the outward normals for these two planes are different.

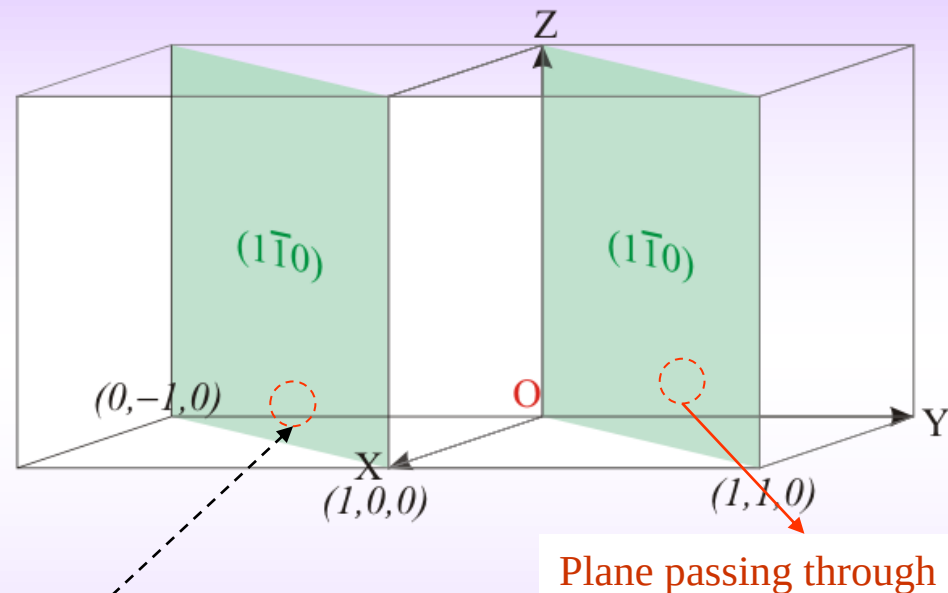
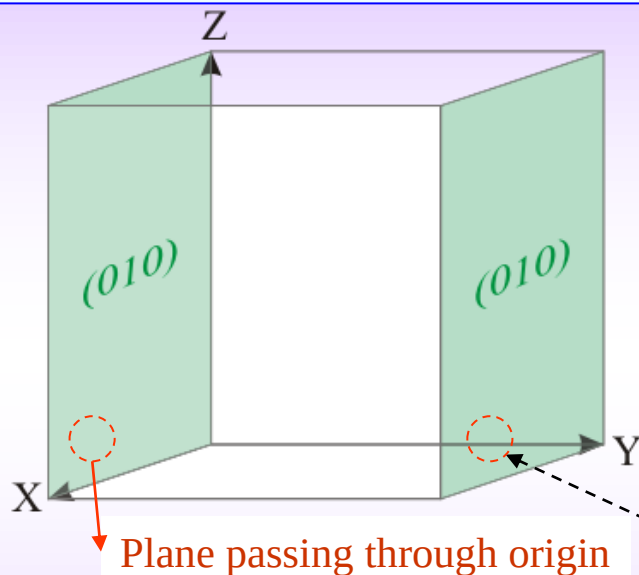
Sometimes, it is also used for a specific plane.

❑ Are the members of the family of {100} planes: (100), (010), (001), (−100), (0−10), (00−1)?

➤ This is a meaningless question without specifying the symmetry of the crystal. The above is true if one is referring to a crystal with (say) $\frac{4}{m} \bar{3} \frac{2}{m}$ symmetry. A ‘family’ is a symmetrically related set (*except for translational symmetry— which is anyhow part of the symbol (100)*).

What about the plane passing through the origin?

- The Miller indices refer to a whole set of parallel planes passing through lattice points. This implies that one of the planes of the infinite set actually passes through the origin.
- As shown below, usually we do not take that particular plane for the determination of Miller indices; but a parallel one, which does not pass through the origin.



Intercepts $\rightarrow \infty 0 \infty$
Plane $\rightarrow (0 \infty 0)$

Hence use this plane

Intercepts $\rightarrow 0 0 \infty$
Plane $\rightarrow (\infty \infty 0)$

We want to avoid infinities in Miller indices

In such cases the plane is translated by a unit distance along the non zero axis/axes and the Miller indices are computed

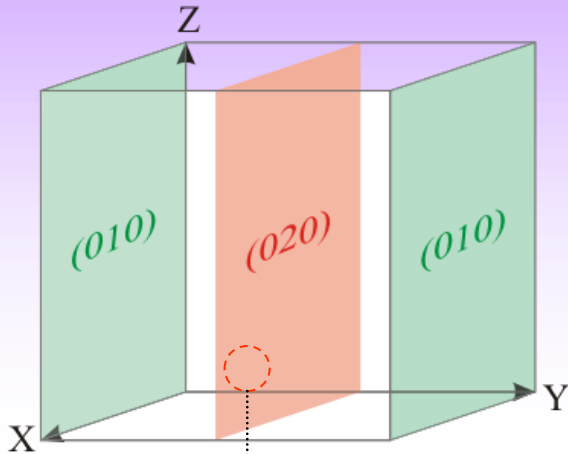
Intercepts $\rightarrow \infty 1 \infty$
Plane $\rightarrow (0 1 0)$

Intercepts $\rightarrow 1 -1 \infty$
Plane $\rightarrow (1 -1 0)$



Funda Check

❑ What about planes passing through fractional lattice spacings?
(We will deal with such fractional intersections with axes in X-ray diffraction).



Intercepts $\rightarrow \infty \frac{1}{2} \infty$

Plane $\rightarrow (0 \ 2 \ 0)$

Note: in Simple cubic lattice this (i.e. every alternate plane in the set) plane will **not** pass through lattice points!!
But then lattice planes have to pass through lattice points!

Why do we consider such planes? We will stumble upon the answer later.

This is an extended use of Miller indices.

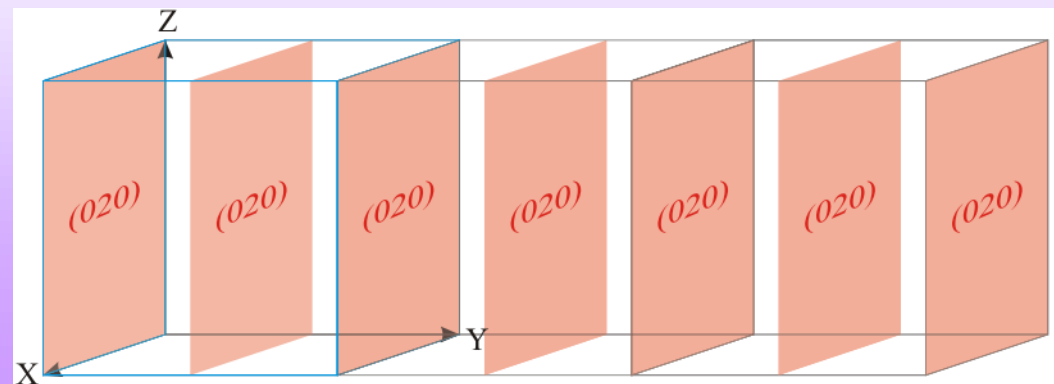
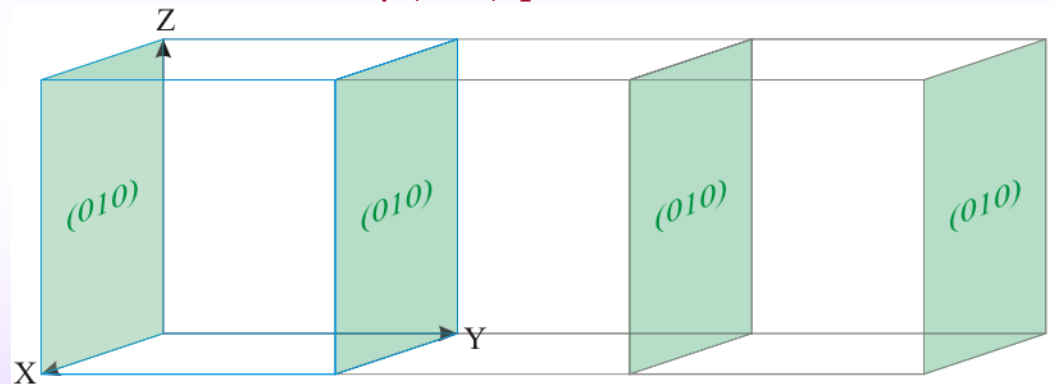
$$d_{010}^{\text{cubic lattice}} = \frac{a}{\sqrt{0^2 + 1^2 + 0^2}} = a$$

$$d_{020}^{\text{cubic lattice}} = \frac{a}{\sqrt{0^2 + 2^2 + 0^2}} = \frac{a}{2}$$

$$d_{020} = \frac{d_{010}}{2}$$

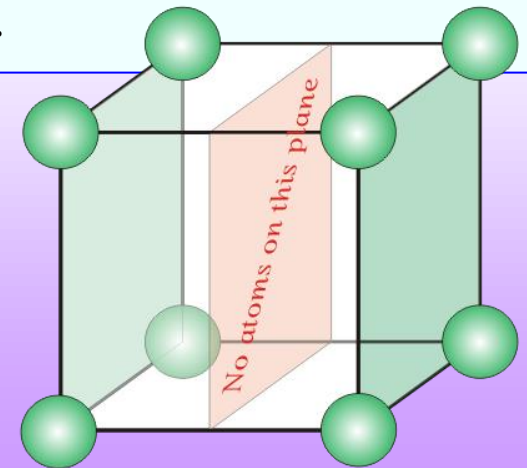
(020) has half the spacing as (010) planes

Actually (020) is a superset of planes as compared to the set of (010) planes



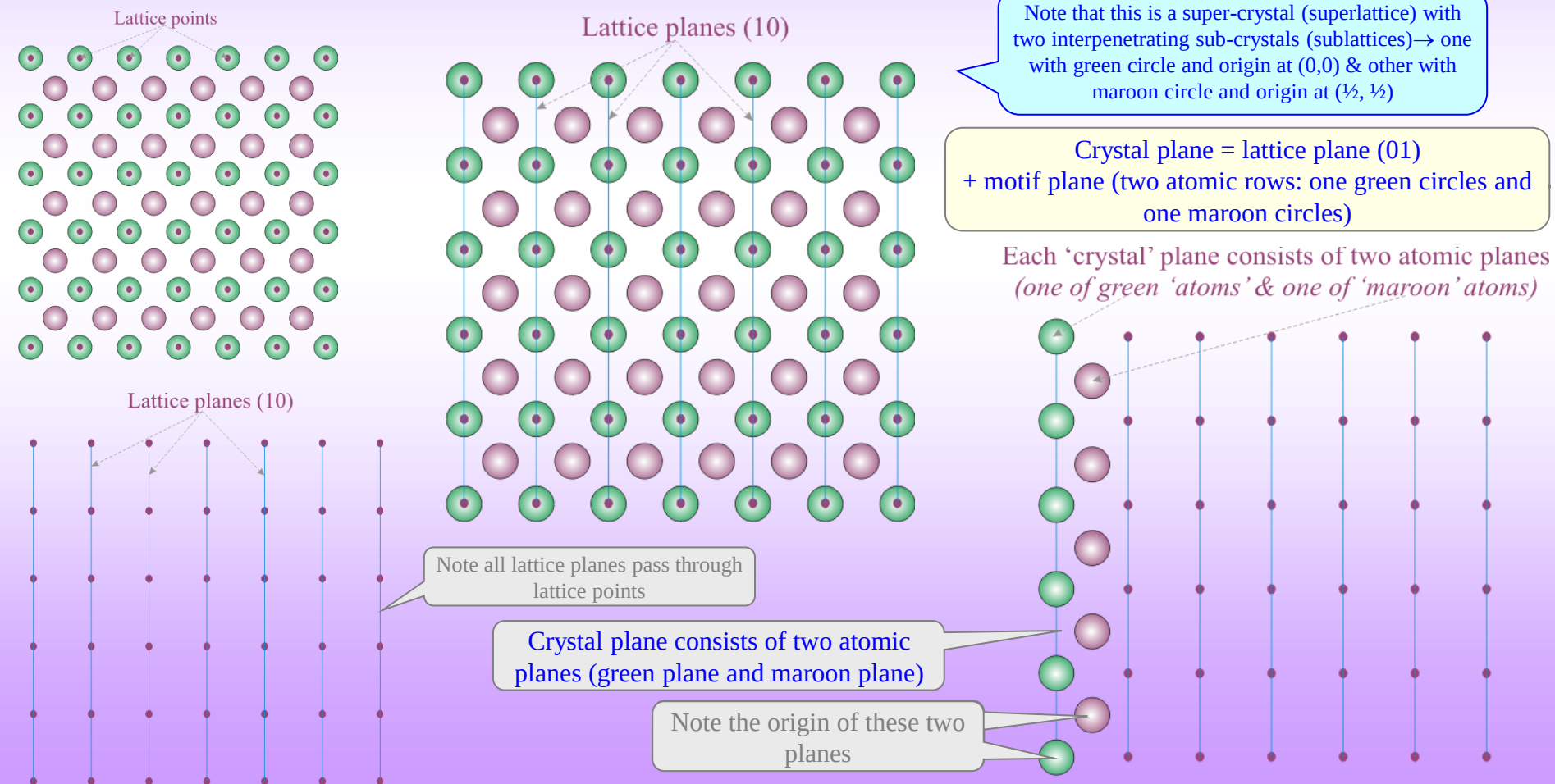
❑ Why talk about (020) planes? Isn't this the same as (010) planes as we factor out common factors in Miller indices?

- ❑ Yes, in Miller indices we usually factor out the common factors.
- ❑ Suppose we consider a simple cubic crystal, *then alternate (020) planes will not have any atoms in them! (And this plane will not pass through lattice points as planes are usually required to do).*
- ❑ Later, when we talk about x-ray diffraction then second order 'reflection' from (010) planes are often considered as first order reflection from (020) planes. This is (one of) the reason we need to consider (020) {or for that matter $(222) \equiv 2(111)$, (333), (220)} kind of planes.
- ❑ Similarly we will also talk about $\frac{1}{2}[110]$ kind of directions. The $\frac{1}{2}$ in front is left out to emphasize the length of the vector (given by the direction). I.e. we are *not only* concerned about a direction, but also the length represented by the vector.



❑ In the crystal below what does the (10) plane contain? Using an 2D example of a crystal.

- ❑ The **‘Crystal’ plane (10)** can be thought of consisting of ‘Lattice’ plane (10) + ‘Motif’ plane (10). I.e. the (10) crystal plane consists of two atomic planes associated with each lattice plane. The green plane and the maroon plane.
- ❑ This concept can be found not only in the superlattice example give below, but also in other crystals. E.g. in the CCP Cu crystal (110) crystal plane consists of two atomic planes of Cu.





Funda Check

❑ Why do we need 3 indices (say for direction) in 3-dimensions?

❑ A direction in 3D can be specified by three angles- or the three direction cosines.

❑ There is one equation connecting the three direction cosines:

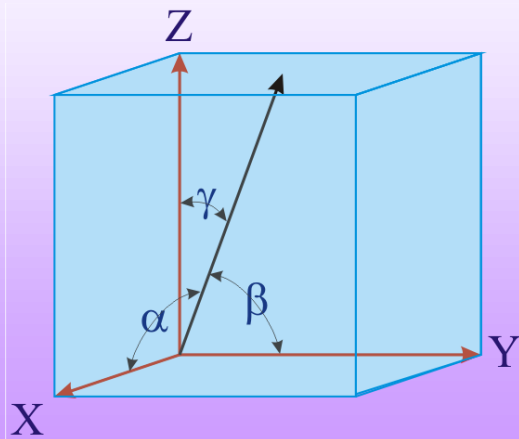
$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

❑ This implies that we required only two independent parameters to describe a direction.
Then why do we need three Miller indices?

❑ The Miller indices prescribe the direction as a vector having a particular length (i.e. this prescription of length requires the additional index)

❑ Similarly three Miller indices are used for a plane (hkl) as this has additional information regarding interplanar spacing. E.g.:

$$d_{hkl}^{\text{cubic lattice}} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$



$$d_{100} = a$$

$$d_{110} = a / \sqrt{2} = a\sqrt{2} / 2$$

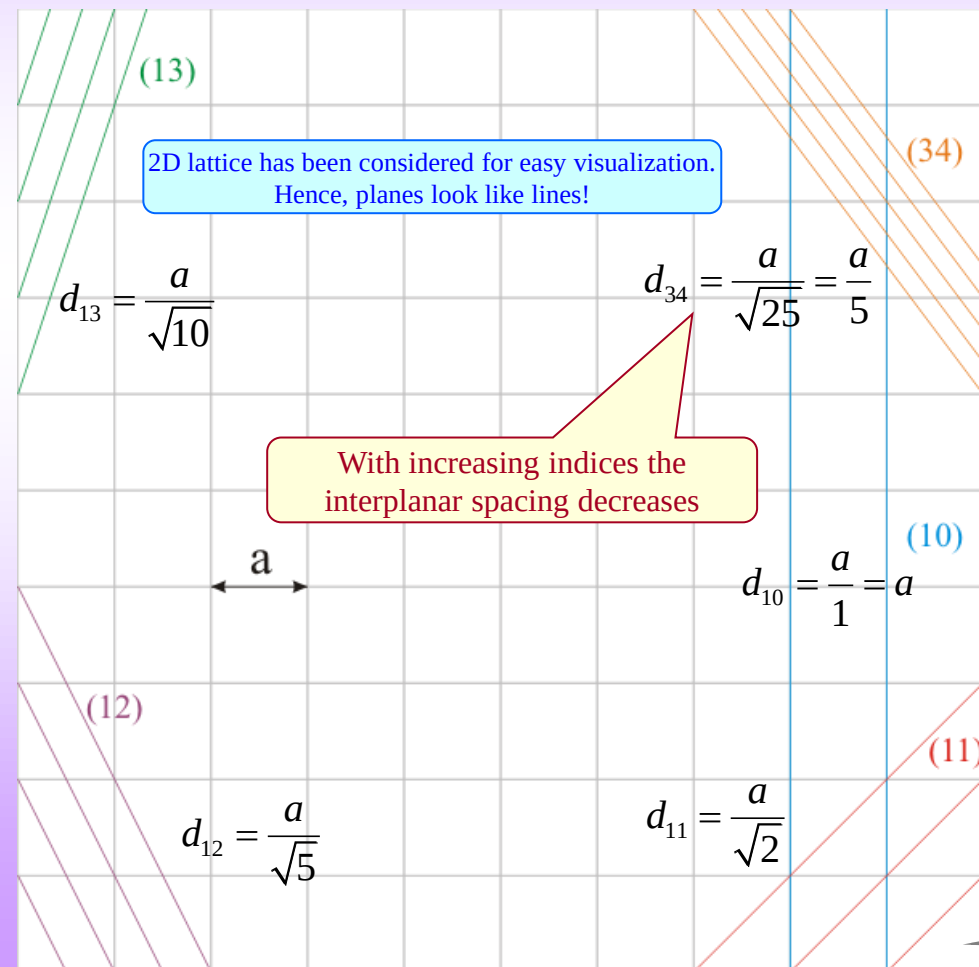
$$d_{111} = a / \sqrt{3} = a\sqrt{3} / 3$$



Funda Check

- 1) What happens to d_{hkl} with increasing hkl ?
- 2) Can planes have spacing less than inter-atomic spacings?
- 3) What happens to lattice density (no. of lattice points per unit area of plane)?
- 4) What is meant by the phrase: 'planes are imaginary'?

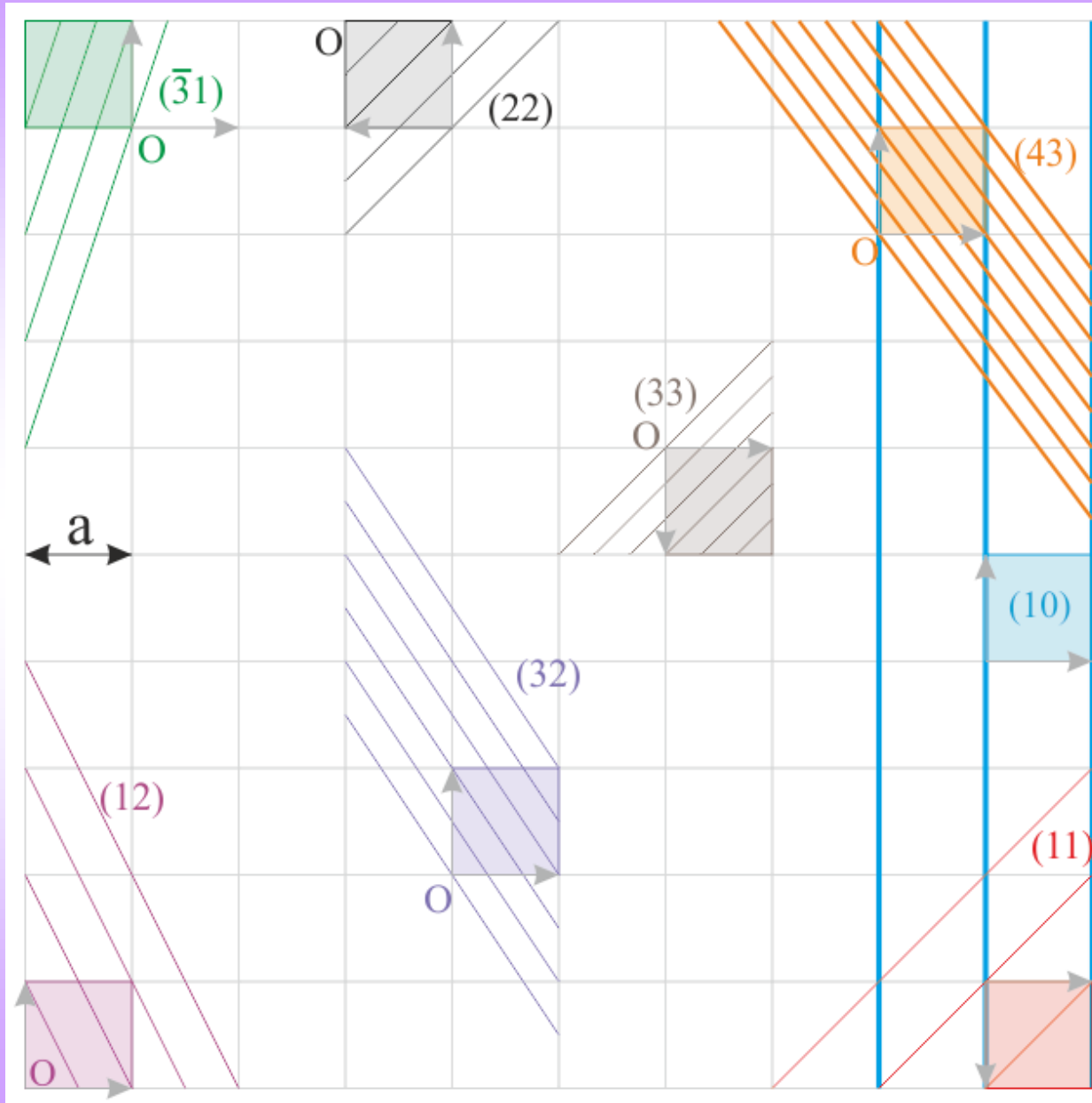
- 1) As h,k,l increases, 'd' decreases $\Rightarrow$ we could have planes with *infinitesimal* spacing.
- 2) The above implies that *inter-planar spacing could be much less than inter-atomic spacing.*



- 3) With increasing indices (h,k,l) the lattice density (or even motif density) decreases.
(in 2D lattice density is measured as no. of lattice points per unit length).
 - E.g. the (10) plane has 1 lattice point for length 'a', while the (11) plane has 1 lattice point for length $a\sqrt{2}$ (i.e. lower density).
- 4) Since we can *draw any number of planes* through the same lattice (as in the figure), clearly the concept of a lattice plane (or for that matter a crystal plane or a lattice direction) is a 'mental' construct (imaginary).

Note: the grey lines do not mean anything (consider this to be a square lattice)

1 more view with more planes and unit cell overlaid



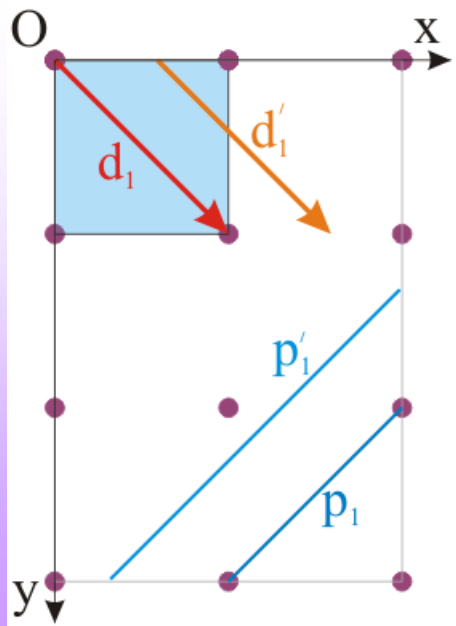
In an upcoming slide we will see how a (hkl) plane will divide the edge, face diagonal and body diagonal of the unit cell

In this 2D version you can already note that diagonal is divided into (h + k) parts

Note the axis is changing locally!

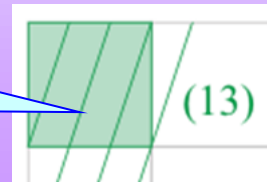
❑ Do planes and directions have to pass through lattice points?

- ❑ In the figure below a direction and plane are marked.
- ❑ In principle d_1 and d'_1 are identical vectorally- but they are positioned differently w.r.t to the origin.
- ❑ Similarly planes p_1 and p'_1 are identical except that they are positioned differently w.r.t to the coordinate axes.
- ❑ In crystallography we usually use d_1 and p_1 (*those which pass through lattice points*) and do not allow any parallel translations (which leads to a situation where these do not pass through lattice points).
- ❑ We have noted earlier that Miller indices (say for planes) contains information about the interplanar spacing and hence the convention.

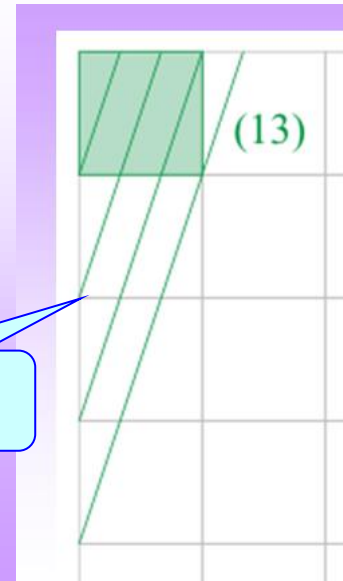


- Sometimes it may seem to us that a given plane or direction is not passing through lattice points, if we consider the part within the unit cell only. E.g. the green planes (13) considered previously.
- In such cases (where actually an intersection occurs, but not seen) we should extend the planes to see the intersection.

Seems like this green plane is not intersecting a lattice point



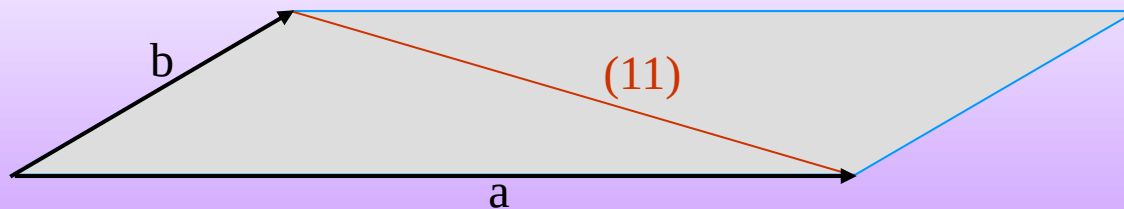
Extend to see the intersection





❑ For a plane (11) what are the *units* of the intercepts?

- ❑ Here we illustrate the concept involved using the (11) plane, but can be applied equally well to directions as well.
- ❑ The (11) plane has intercepts along the crystallographic axis at (1,0) and (0,1).
- ❑ In a given lattice/crystal the 'a' and 'b' axis need not be of equal length (further they may be inclined to each other). This implies that though the intercepts are one unit along 'a' and 'b', their physical lengths may be very different (as in the figure below).



(111)

Further points about (111) planes

Family of {111} planes within the cubic unit cell (Light green triangle and light blue triangle are (111) planes within the unit cell).

The Orange hexagon is parallel to these planes.

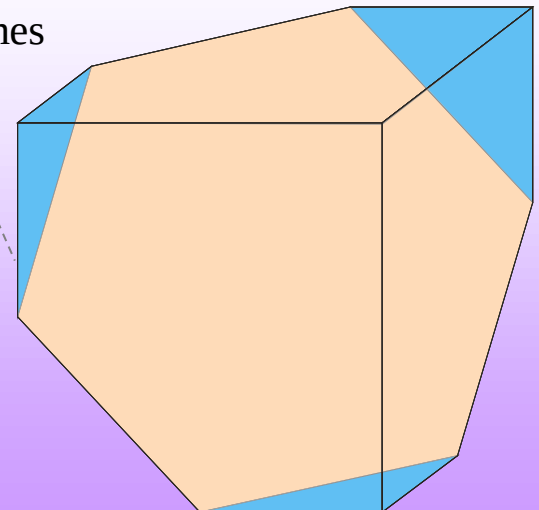
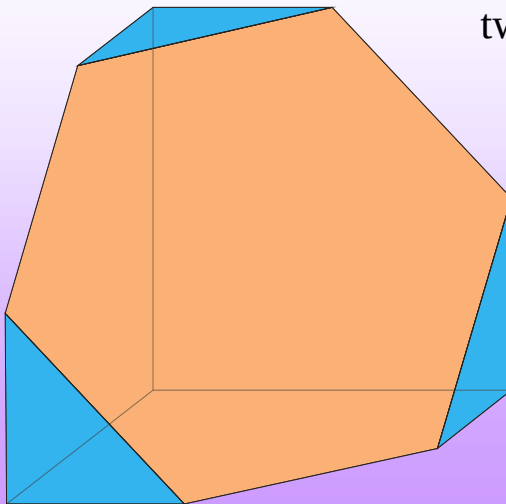
$$d_{(111)}^{cubic} = \frac{a}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{a}{\sqrt{3}} = a \frac{\sqrt{3}}{3} = \frac{\text{Body diagonal length}}{3}$$

Orange plane NOT part of (111) set (does not pass through lattice points- also see next slide)

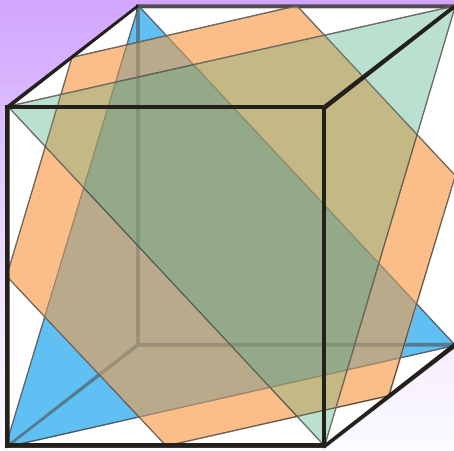
Blue and green planes are (111)

The (111) plane(s) {the blue and the orange ones} trisects the body diagonal

The Orange hexagon Plane cuts the cube into two polyhedra of equal volumes



The central (111) plane (orange colour) is not a 'space filling' plane!

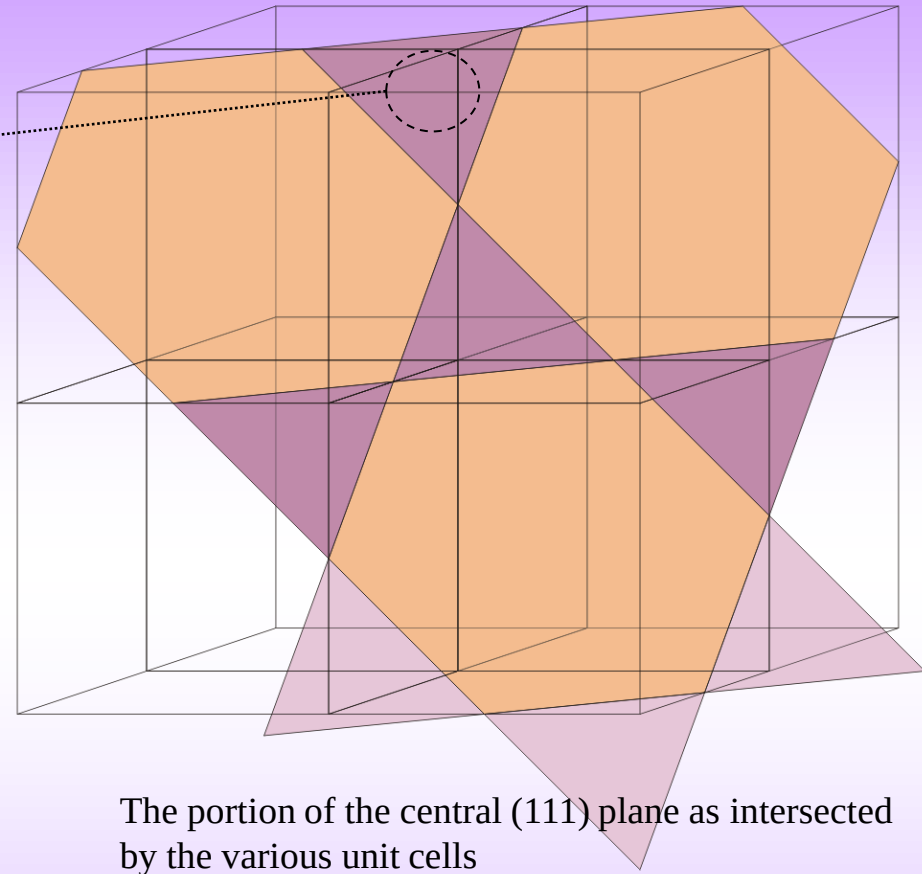


Portion of the (111) plane not included within the unit cell

Suppose we want to make a calculation of areal density (area fraction occupied by atoms) of atoms on the (111) plane- e.g. for a BCC crystal.

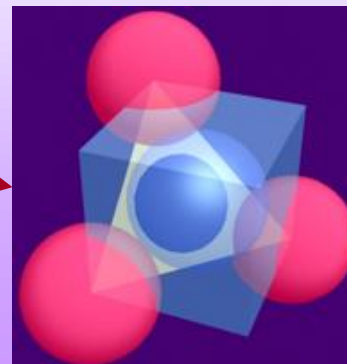
Q) Can any of these (111) planes be used for the calculation?

A) If the calculation is being done within the unit cell then the central orange plane cannot be used as it (the hexagonal portion) is *not* space filling → as shown in the figure on the right.



The portion of the central (111) plane as intersected by the various unit cells

Solved Example What is the true areal fraction of atoms lying in the (111) plane of a BCC crystal?



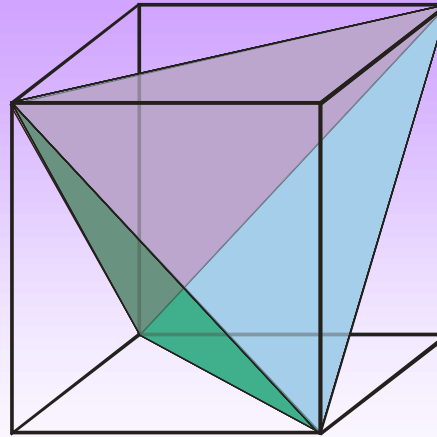
Video: (111) plane in BCC crystal

Low resolution

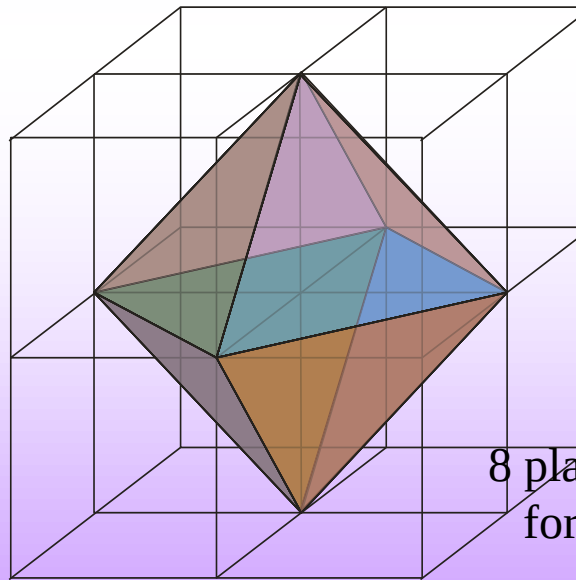
Video: (111) plane in BCC crystal

Members of a family of planes in cubic crystal/lattice

Index	n^*	
$\{100\}$	6	
$\{110\}$	12	The (110) plane bisects the face diagonal
$\{111\}$	8	The (111) plane trisects the body diagonal
$\{210\}$	24	
$\{211\}$	24	
$\{221\}$	24	
$\{310\}$	24	
$\{311\}$	24	
$\{320\}$	24	
$\{321\}$	48	



Tetrahedron inscribed inside a cube with bounding planes belonging to the $\{111\}_{\text{cubic lattice}}$ family (subset of the full family)



8 planes of $\{111\}_{\text{cubic lattice}}$ family forming a regular octahedron

n^* is the No. of members in a cubic lattice

Summary of notations

A family is also referred to as a *symmetrical set*

	Symbol		Alternate symbols	
Direction	[]	[uvw]		Particular direction
	< >	<uvw>	[[]]	Family of directions
Plane	()	(hkl)		Particular plane
	{ }	{hkl}	(())	Family of planes
Point	..	.xyz.	[[]]	Particular point
	::	:xyz:		Family of points

Points about (hkl) planes

For a set of translationally equivalent lattice planes will divide:

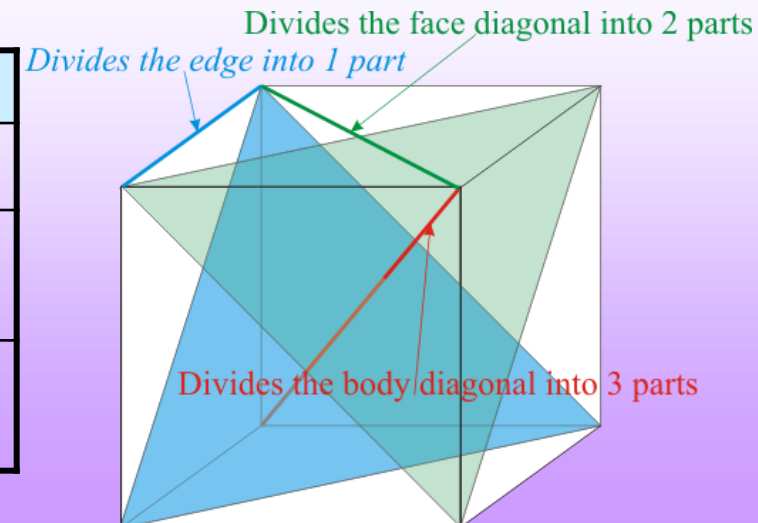
Entity being divided (Dimension containing the entity)		Direction	Number of parts
Cell edge (1D)	a	[100]	h
	b	[010]	k
	c	[001]	l
Diagonal of cell face (2D)	(100)	[011]	(k + l)
	(010)	[101]	(l + h)
	(001)	[110]	(h + k)
Body diagonal (3D)		[111]	(h + k + l)

This implies that the (111) planes will divide the face diagonals into two parts and the body diagonal into 3 parts.



Some general points

Condition	(hkl) will pass through
h even	midpoint of 'a'
(k + l) even	face centre (001) midpoint of face diagonal (001)
(h + k + l) even	body centre midpoint of body diagonal



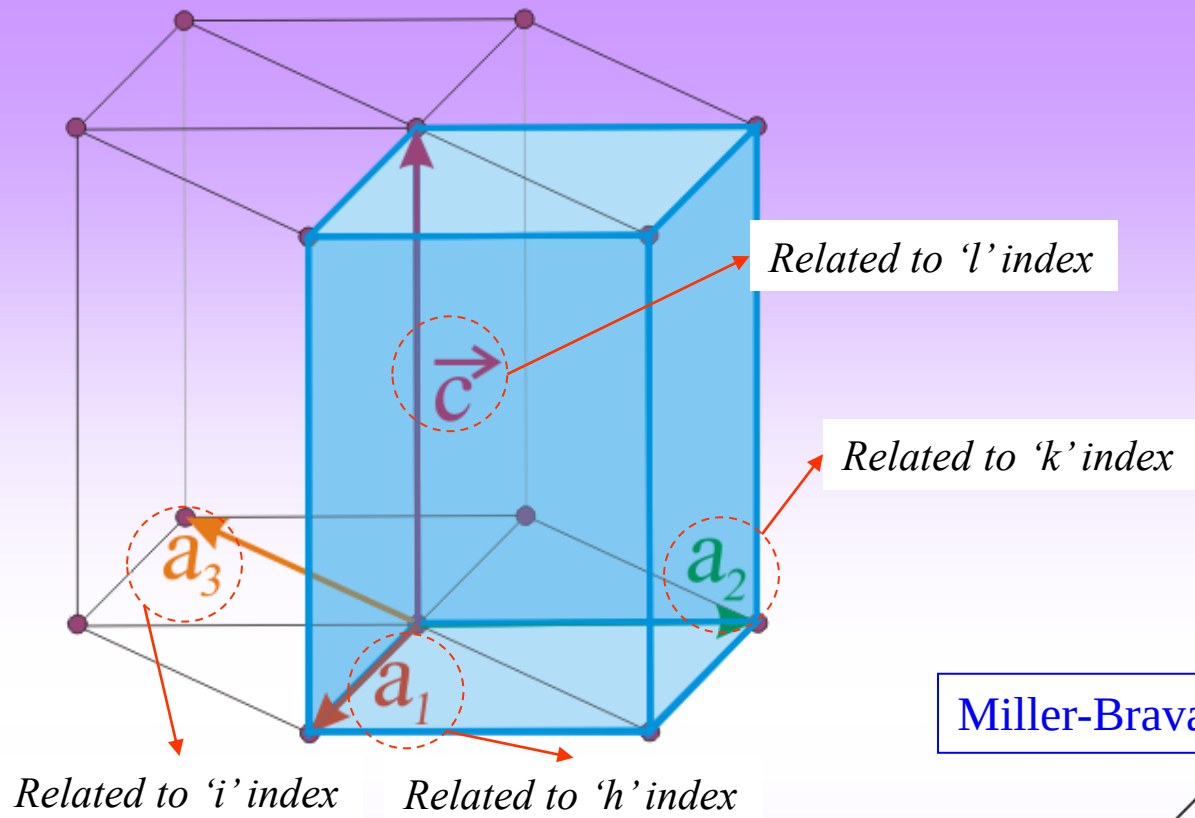
- Let the 'c' axis be the unique axis for the tetragonal lattice. The tetragonal lattice has the highest symmetry possible for the tetragonal system ($4/m\ 2/m\ 2/m$).
- The family of planes can be written as $\{1\ 1\ 0\}$. The first two indices can be permuted, while the third one cannot.
- Hence, $\{1\ 1\ 0\} \equiv (110), (-110), (1\bar{1}0), (\bar{1}\bar{1}0)$.
- As we know $(\bar{1}\bar{1}0) \equiv -(110) \equiv (110)$. I.e. all these planes belong to the same set.

Hexagonal crystals → The Miller-Bravais Indices

- ❑ Directions and planes in hexagonal lattices and crystals are designated by the **4-index** Miller-Bravais notation. *Note: in 3D we need only 3 indices.*
- ❑ The Miller-Bravais notation can be a little tricky to learn.
- ❑ In the four index notation the following points are to be noted.
 - The first three indices are a **symmetrically** related set on the basal plane.
 - The third index is a *redundant one* (which can be derived from the first two as in the formula: $i = -(h+k)$) and is introduced to make sure that members of a family of directions or planes have a set of numbers which are identical.
 - This is because in 2D two indices *suffice* to describe a lattice (or crystal).
 - The fourth index represents the 'c' axis ($\perp$ to the basal plane).
- ❑ Hence the first three indices in a hexagonal lattice can be permuted to get the different members of a family; while, the fourth index is kept separate.

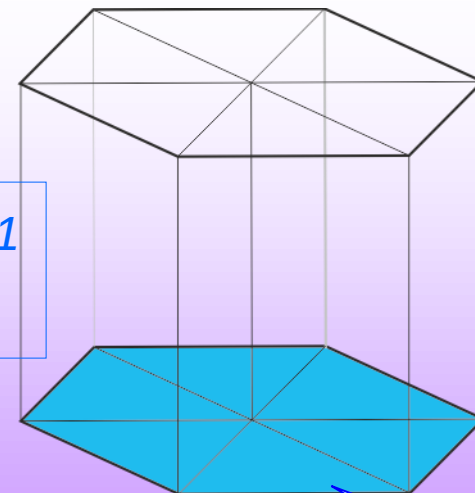
$$\begin{matrix} (h & k & i & l) \\ i = -(h + k) \end{matrix}$$

- ❑ There is only one hexagonal lattice (in 3D) and this has 6-fold symmetry.
- ❑ Hexagonal crystals on the other hand may or may not have pure 6-fold symmetry. However, they do have some form of a '6' symmetry (roto-inversion or screw: $-6, 6_1, 6_2, 6_3$).
- ❑ The hexagonal basis set is used to describe all hexagonal lattices and crystals.



Miller-Bravais Indices for the Basal Plane

Intercepts $\rightarrow \infty \infty \infty 1$
 Plane $\rightarrow (0 \ 0 \ 0 \ 1)$



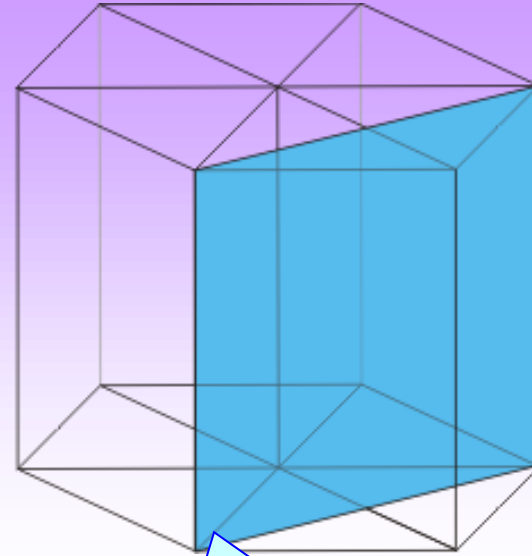
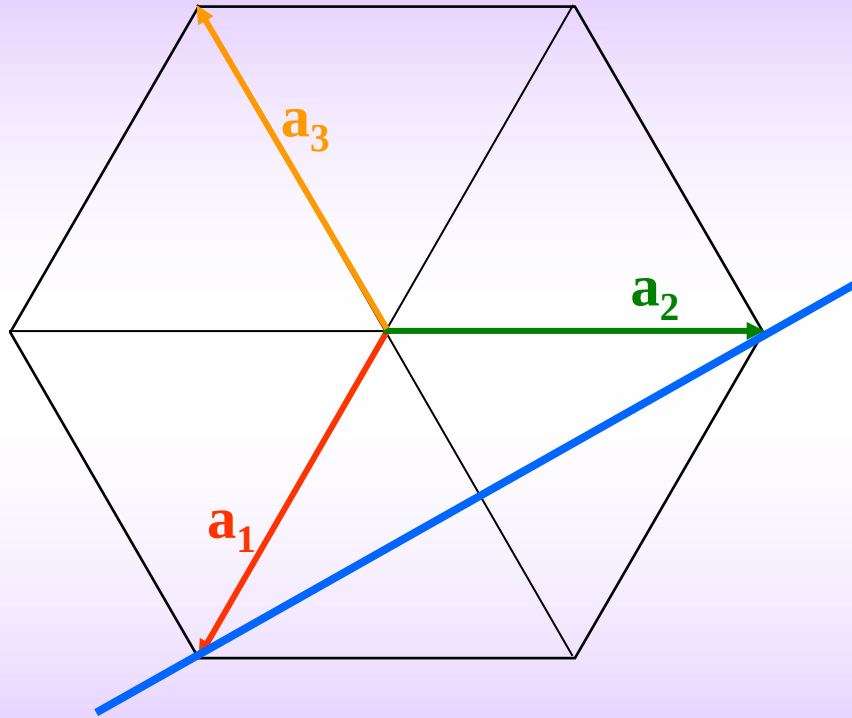
Basal Plane

Intercepts $\rightarrow 1 \ 1 \ -\frac{1}{2} \ \infty$

Plane $\rightarrow (1 \ 1 \ \bar{2} \ 0)$

$(h \ k \ i \ l)$

$i = -(h + k)$

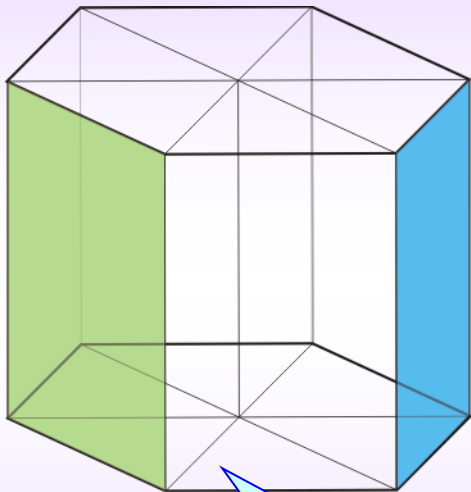
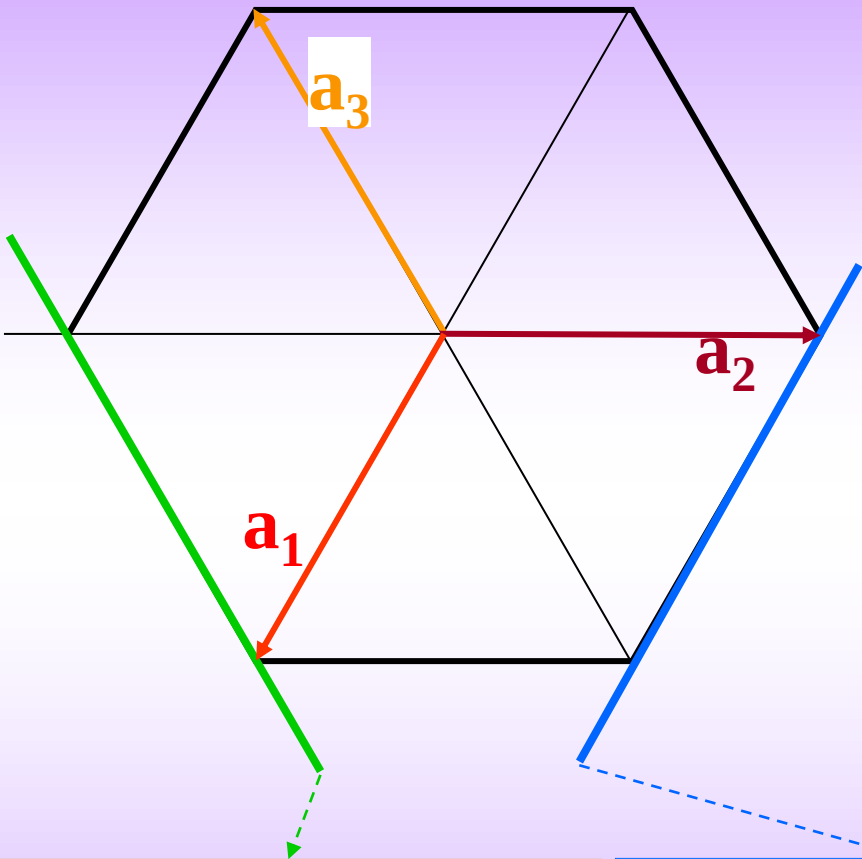


Planes which have ∞ intercept along c-axis (i.e. vertical planes) are called
Prism planes

The use of the 4 index notation is to bring out the equivalence between crystallographically equivalent planes and directions (as will become clear in coming slides)

Examples to show the utility of the 4 index notation

Obviously (related by 3-fold symmetry), the ‘green’ and ‘blue’ planes belong to the same family and first three indices have the same set of numbers (as brought out by the Miller-Bravais system)

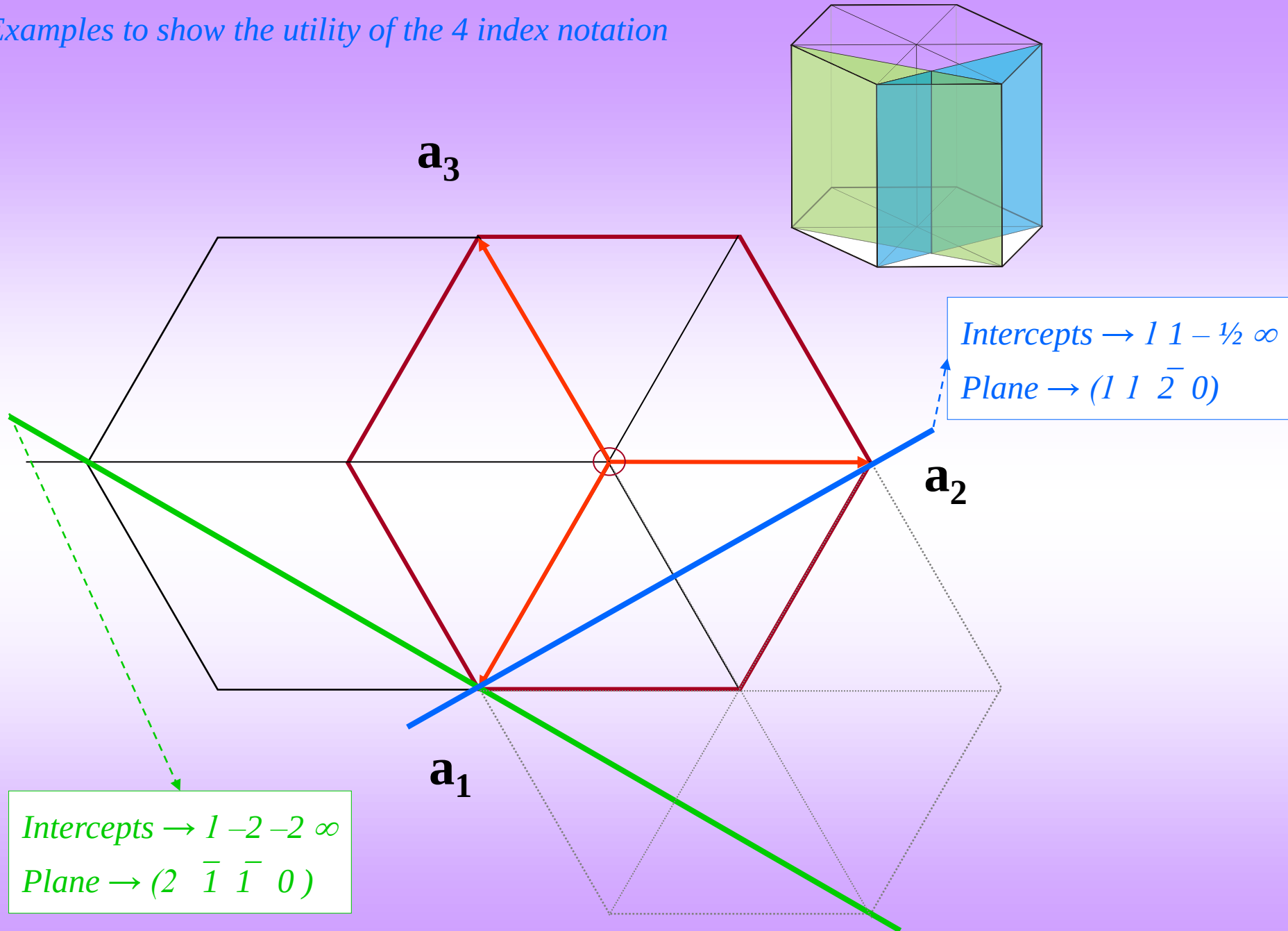


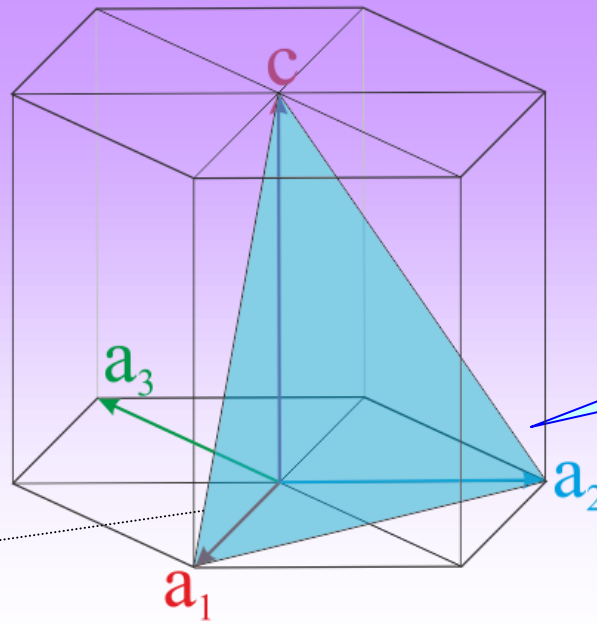
Planes which have ∞ intercept along c-axis (i.e. vertical planes) are called Prism planes

Intercepts $\rightarrow 1 \ -1 \ \infty \ \infty$
 Miller (3 index) $\rightarrow (1 \ \bar{1} \ _0)$
 Miller-Bravais $\rightarrow (1 \ \bar{1} \ 0 \ 0)$

Intercepts $\rightarrow \infty \ 1 \ -1 \ \infty$
 Miller (3 index) $\rightarrow (0 \ 1 \ _0)$
 Miller-Bravais $\rightarrow (0 \ 1 \ \bar{1} \ 0)$

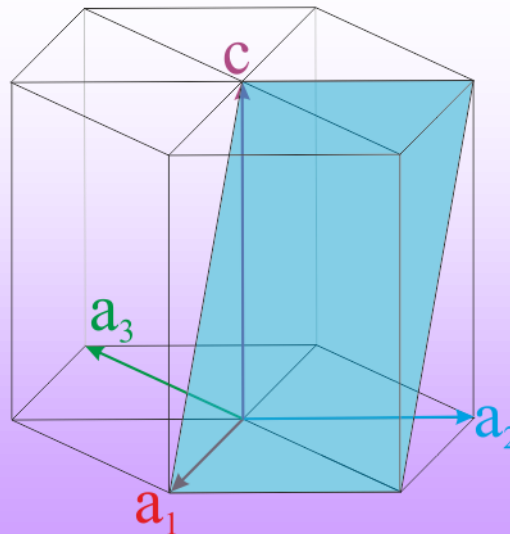
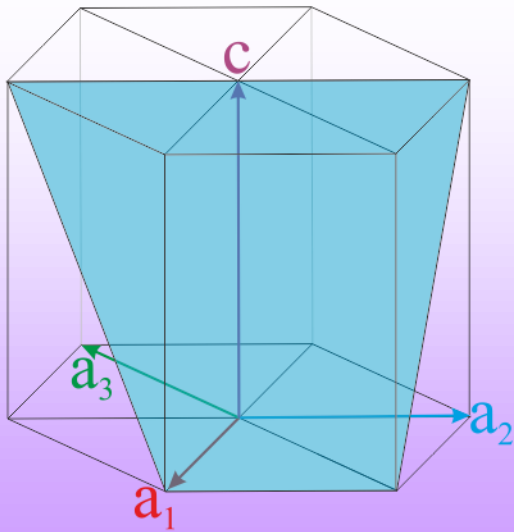
Examples to show the utility of the 4 index notation





Inclined planes which have finite intercept along c-axis are called **Pyramidal planes**

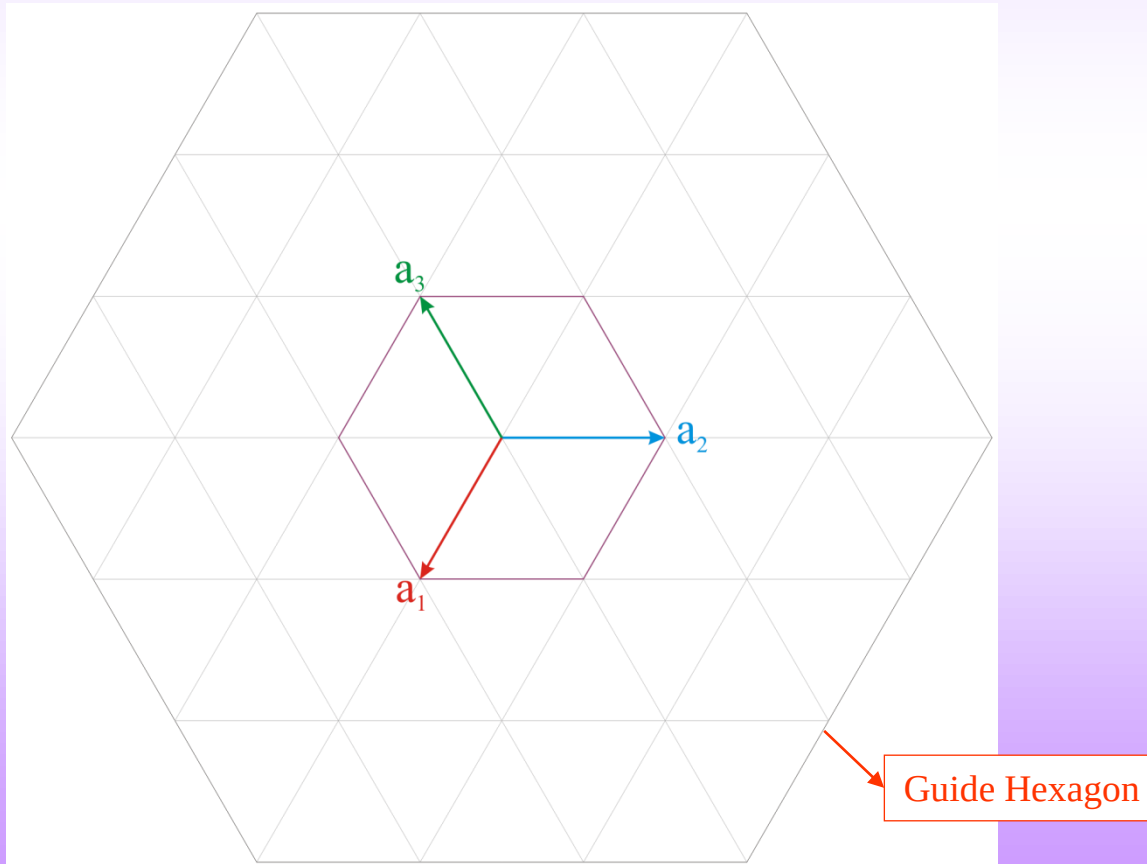
Intercepts $\rightarrow 1 \ 1 \ -\frac{1}{2} \ 1$
Plane $\rightarrow (1 \ 1 \ \bar{2} \ 1)$



Intercepts $\rightarrow 1 \ \infty \ -1 \ 1$
Plane $\rightarrow (1 \ 0 \ \bar{1} \ 1)$

Directions

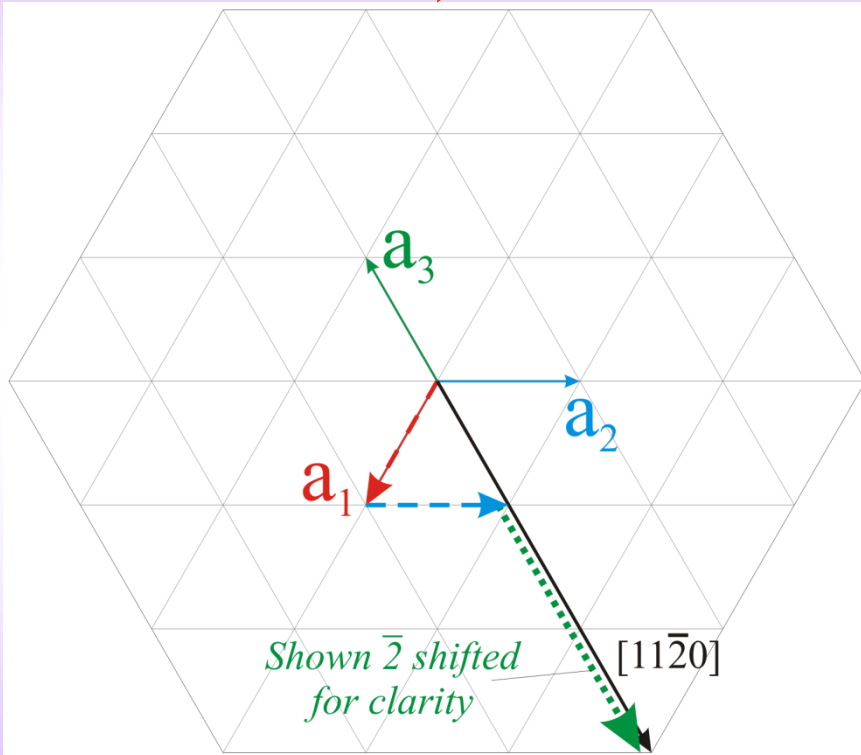
- ❑ One has to be careful in determining directions in the Miller-Bravais system.
- ❑ Basis vectors a_1 , a_2 & a_3 are symmetrically related by a six fold axis.
- ❑ The 3rd index is redundant and is included to bring out the equality between equivalent directions (like in the case of planes).
- ❑ In the drawing of the directions we use an additional guide hexagon 3 times the unit basis vectors (a_i).



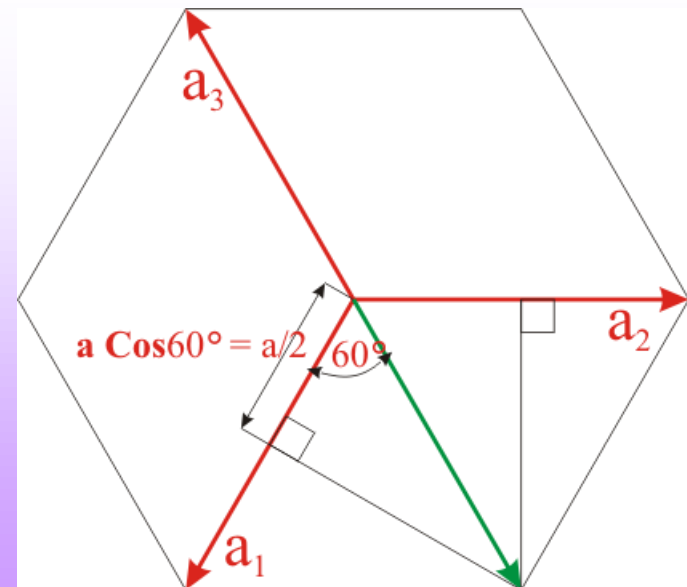
Directions

Drawing the $[11\bar{2}0]$ direction

- Trace a path along the basis vectors as required by the direction. In the current example move 1 unit along a_1 , 1 unit along a_2 and -2 units along a_3 .
- Directions are projected onto the basis vectors to determine the components and hence the Miller-Bravais indices can be determined as in the table.

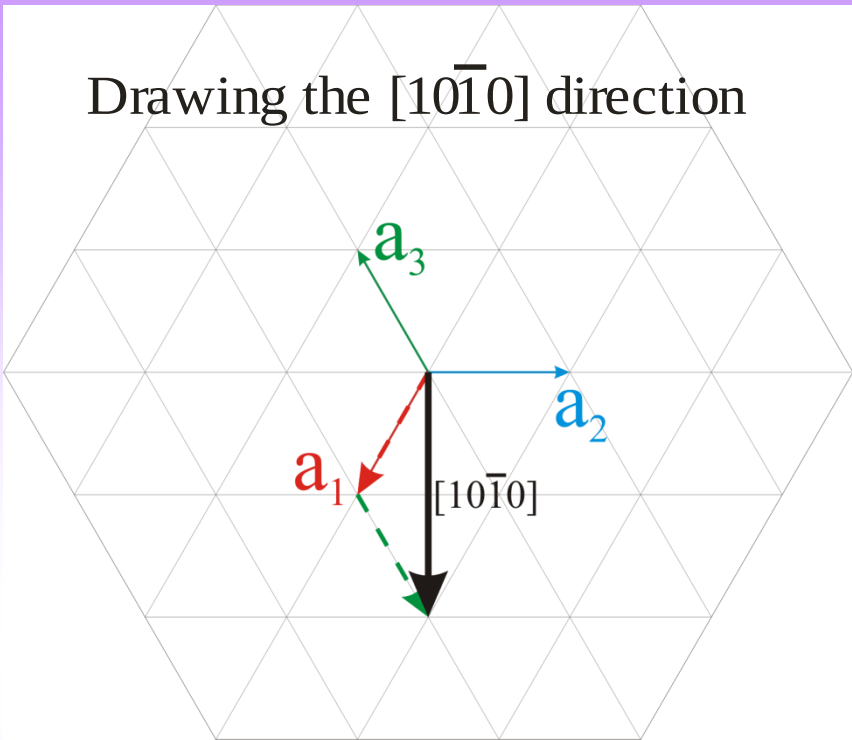


	a_1	a_2	a_3
Projections	$a/2$	$a/2$	$-a$
Normalized wrt LP	$1/2$	$1/2$	-1
Factorization	1	1	-2
Indices	$[1\ 1\ \bar{2}\ 0]$		



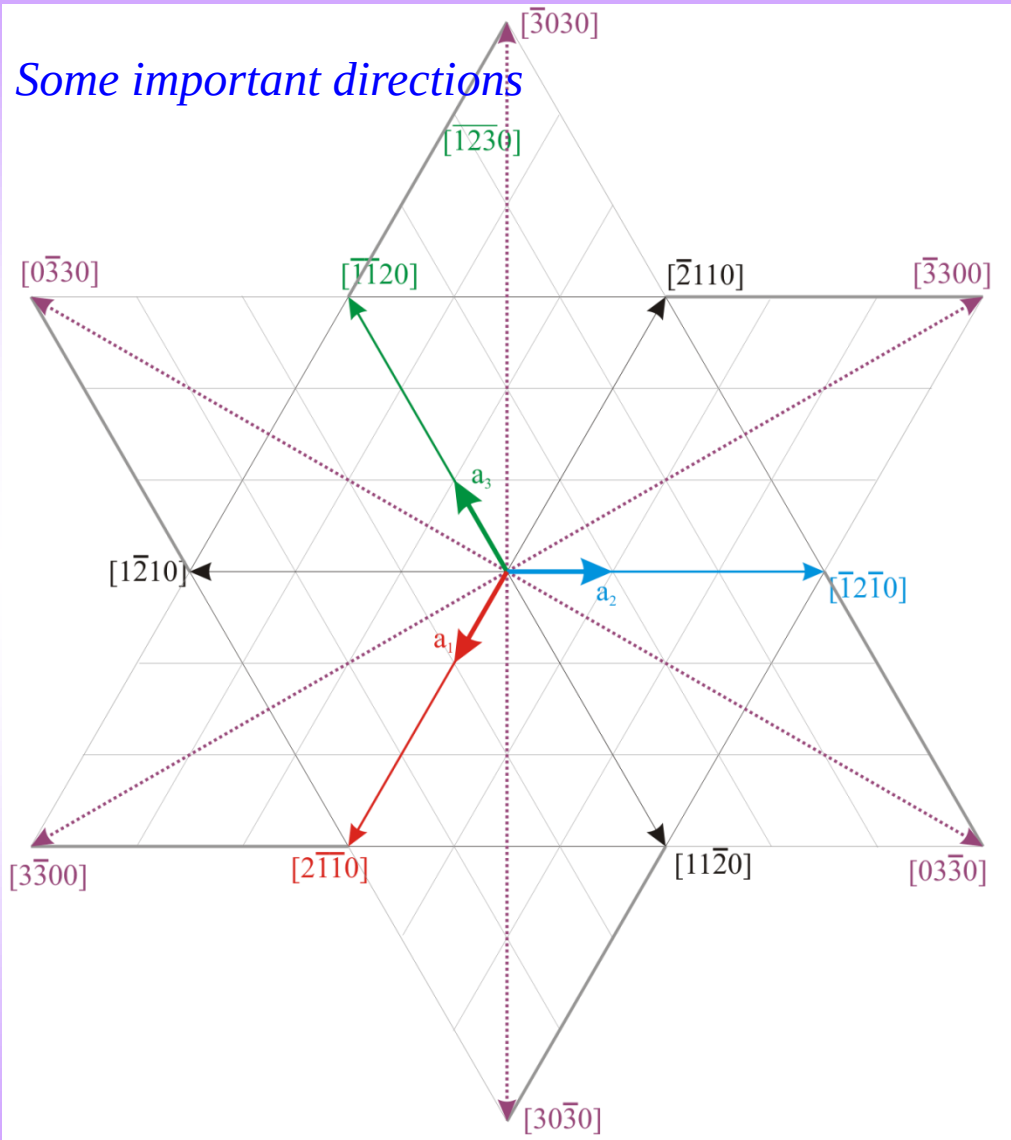
We do similar exercises to draw other directions as well

Drawing the $[10\bar{1}0]$ direction



	a_1	a_2	a_3
Projections	$3a/2$	0	$-3a/2$
Normalized wrt LP	$3/2$	0	$-3/2$
Factorization	1	0	-1
Indices	$[1\ 0\ -1\ 0]$		

Some important directions



Overlaying planes and directions

- ❑ Note that for planes of the type (000l) or (hki0) are perpendicular to the respective directions [0001] or [hki0] → $(000l) \perp [0001]$, $(hki0) \perp [hki0]$.
- ❑ However, in general (hkil) is not perpendicular to [hkil], except if c/a ratio is $\sqrt{3/2}$.
- ❑ The direction perpendicular to a particular plane will depend on the c/a ratio and may have high indices or even be irrational.

Transformation between 3-index $[UVW]$ and 4-index $[uvw]$ notations

$$U = u - t \quad V = v - t \quad W = w$$

$$u = \frac{1}{3}(2U - V) \quad v = \frac{1}{3}(2V - U)$$

$$t = -(u + v)$$

$$w = W$$

- Directions in the hexagonal system can be expressed in many ways
- 3-indices:
By the three vector components along a_1 , a_2 and c :
$$r_{UVW} = Ua_1 + Va_2 + Wc$$
- In the three index notation equivalent directions may not seem equivalent; while, in the four index notation the equivalence is brought out.

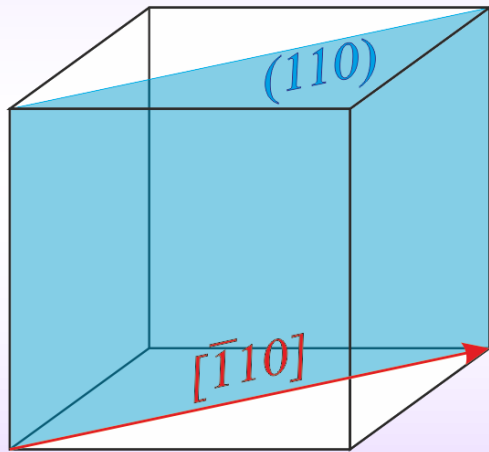
Weiss Zone Law

□ If the Miller plane (hkl) contains (or is parallel to) the direction $[uvw]$ then:

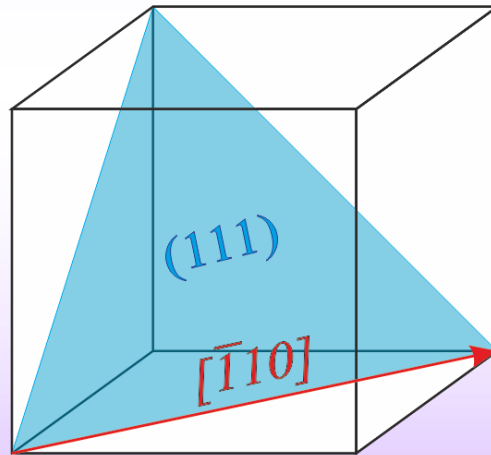
$$h.u + k.v + l.w = 0$$

□ This relation is valid for all crystal systems (referring to the standard unit cell).

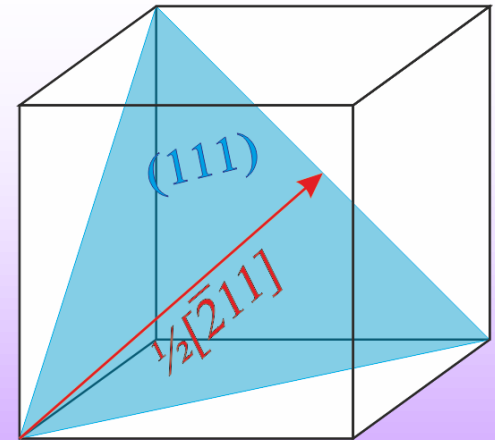
The red directions lie on the blue planes



$$(1 \cdot \bar{1}) + (1 \cdot 1) + (1 \cdot 0) = 0$$



$$(1 \cdot \bar{1}) + (1 \cdot 1) + (0 \cdot 0) = 0$$

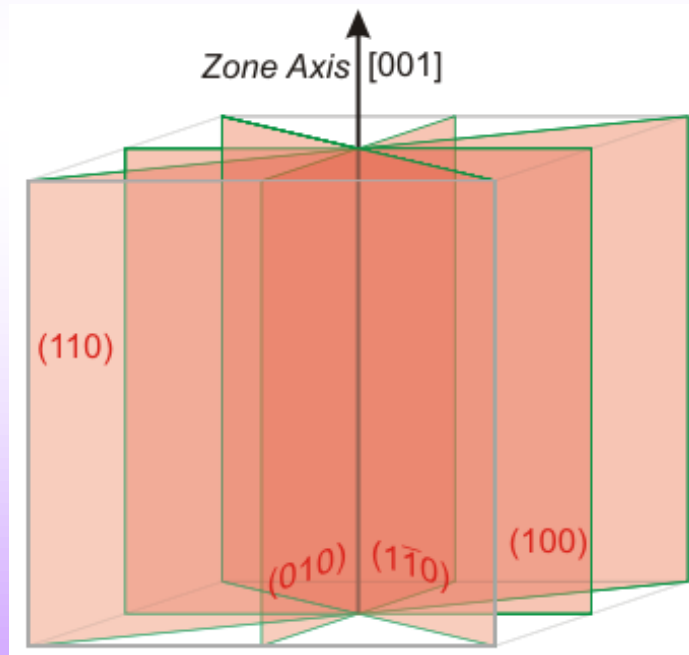


$$\frac{1}{2} \left[(1 \cdot \bar{2}) + (1 \cdot 1) + (1 \cdot 1) \right] = 0$$

**Solved
Example**

Zone Axis

- ❑ The direction common to a set of planes is called the zone axis of those planes.
- ❑ E.g. $[001]$ lies on (110) , $(1\bar{1}0)$, (100) , (210) etc.
- ❑ If $(h_1 \ k_1 \ l_1)$ & $(h_2 \ k_2 \ l_2)$ are two planes having a common direction $[uvw]$ then according to Weiss zone law:
$$u.h_1 + v.k_1 + w.l_1 = 0 \text{ \& \> } u.h_2 + v.k_2 + w.l_2 = 0$$
- ❑ This concept is very useful in Selected Area Diffraction Patterns (SADP) in a TEM.



Note: Planes of a zone lie on a great circle in the stereographic projection

Directions $\perp$ Planes

Many a times (or sometimes if you like) a direction may be perpendicular to the plane represented by the same indices. E.g. in the cubic system, the $[120]$ direction is perpendicular to the (120) plane. As evident from this statement, this is not always true. We list below the cases in which this is true.

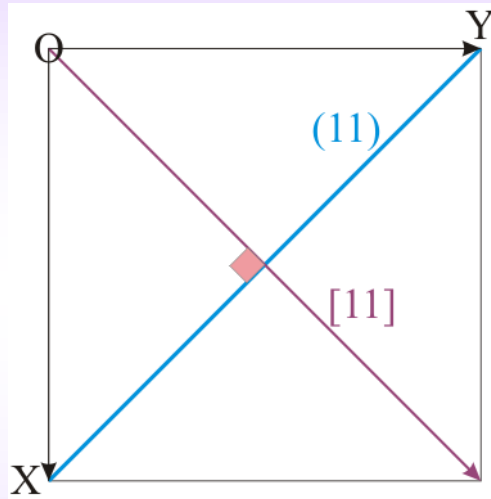
- ❑ Cubic system*: $(hkl) \perp [hkl]$
- ❑ Tetragonal system*: only special planes are $\perp$ to the direction with same indices:
 $[100] \perp (100)$, $[010] \perp (010)$, $[001] \perp (001)$, $[110] \perp (110)$
($[101]$ not $\perp (101)$)
- ❑ Orthorhombic system*:
 $[100] \perp (100)$, $[010] \perp (010)$, $[001] \perp (001)$
- ❑ Hexagonal system*
 - ▶ This is for a general c/a ratio only this is true: $[0001] \perp (0001)$.
 - ▶ For a Hexagonal crystal with the special c/a ratio = $\sqrt{3/2}$ *
→ the cubic rule is followed (i.e. all planes are $\perp$ to all directions).
- ❑ Monoclinic system**: $[010] \perp (010)$
- ❑ Other than these a general $[hkl]$ is **NOT** $\perp (hkl)$

* This is called the Frank's ratio and such crystals with this c/a ratio are called Frank's pseudo-cubic phases.

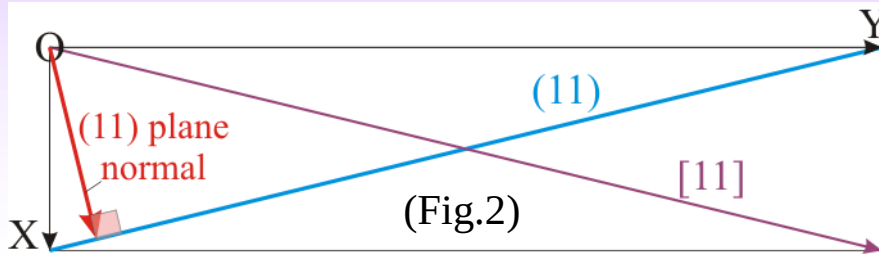
** Here we are referring to the conventional unit cell chosen (e.g. $a=b=c$, $\alpha=\beta=\gamma=90^\circ$ for cubic) and not the symmetry of the crystal.

Which direction is perpendicular to which plane?

- ❑ In the cubic system all directions are perpendicular to the corresponding planes ($(hkl) \perp [hkl]$). 2D example of the same is given in the figure on the left (Fig.1).
- ❑ However, this is not universally true. To visualize this refer to Fig.2 and Fig.3 below.

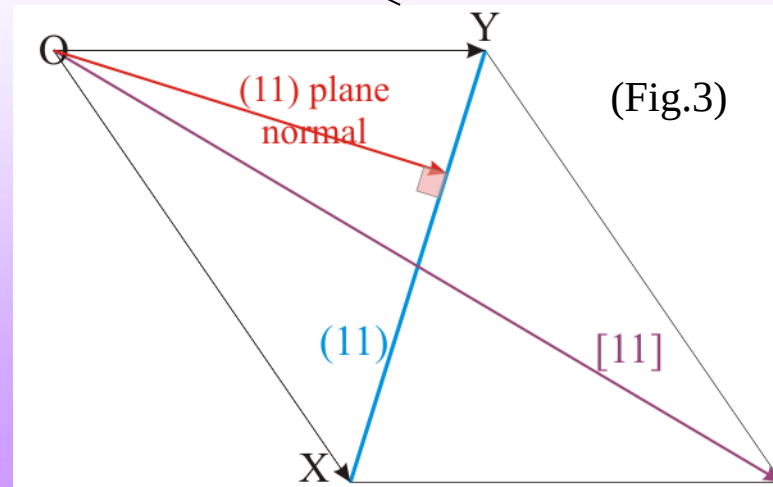


(Fig.1)



(Fig.2)

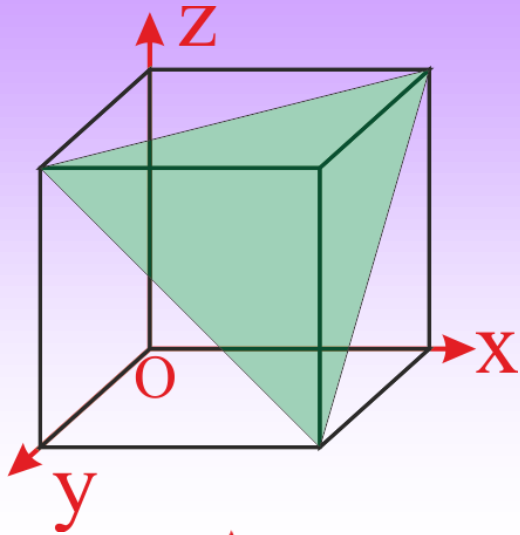
Note that *plane normal to (11) plane* is not the same as the *[11] direction*



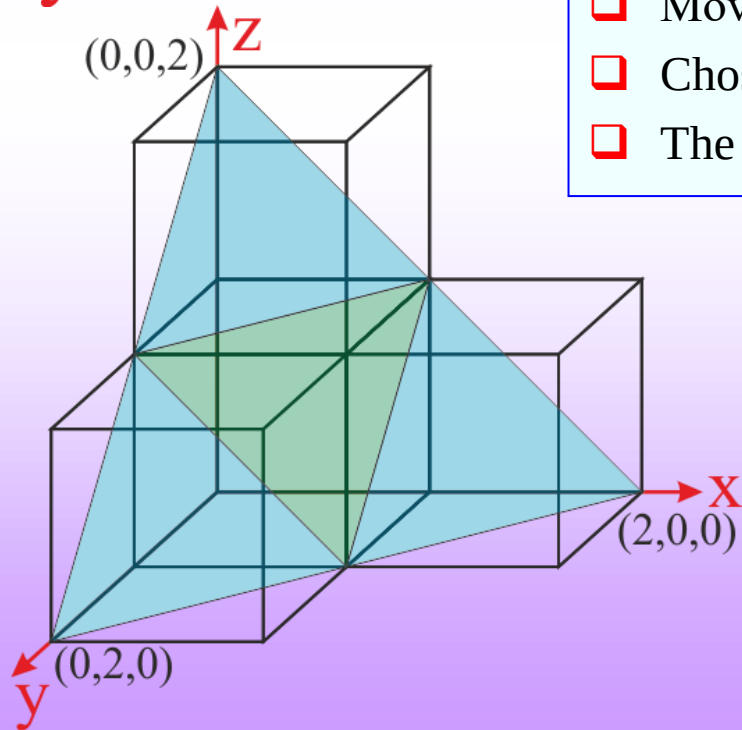
(Fig.3)

Q & A

What are the Miller indices of the green plane in the figure below?



- ☐ Extend the plane to intersect the x,y,z axes.
- ☐ The intercepts are: 2,2,2
- ☐ Reciprocal: $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$
- ☐ Smallest ratio: 1,1,1
- ☐ Enclose in brackets to get Miller indices: (111)



- ☐ Another method.
- ☐ Move origin ('O') to opposite vertex (of the cube).
- ☐ Chose new axes as: $-x$, $-y$, $-z$.
- ☐ The new intercepts will be: 1,1,1

Multiplicity factor

This concept is very useful in [X-Ray Diffraction](#)

Advanced Topic

Cubic	hkl	hhl	hk0	hh0	hhh	h00	
	48*	24	24*	12	8	6	
Hexagonal	hk.l	hh.l	h0.l	hk.0	hh.0	h0.0	00.l
	24*	12*	12*	12*	6	6	2
Tetragonal	hkl	hhl	h0l	hk0	hh0	h00	00l
	16*	8	8	8*	4	4	2
Orthorhombic	hkl	hk0	h0l	0kl	h00	0k0	00l
	8	4	4	4	2	2	2
Monoclinic	hkl	h0l	0k0				
	4	2	2				
Triclinic	hkl						
	2						

* Altered in crystals with lower symmetry (*of the same crystal class*)

Q & A

What are the members of the family $\langle 110 \rangle$ & $\{111\}$ for the point group 23?

- ❑ Note that 23 is a cubic point group (the 3 occurs in the second place) and it lacks a 4 fold and a centre of inversion.
- ❑ The point group has 3-fold operators along $\langle 111 \rangle$ directions and 2-fold along $\langle 100 \rangle$ directions.
- ❑ Action of 2-fold along $[001]$: $[110] \rightarrow [-1-10]$ (Fig.1).
- ❑ Action of 3-fold along $[111]$: $[110]$ & $[-1-10] \rightarrow [011]$, $[101]$ and their negatives (Fig.2).
- ❑ Action of 3-fold along $[-111]$: $[011]$ & $[0-1-1] \rightarrow [1-10]$, $[-101]$ and their negatives (Fig.3).
- ❑ Action of 3-fold along $[1-11]$: $[101]$ & $[-10-1] \rightarrow [01-1]$ and its negative (Fig.4).
- ❑ Hence, $\langle 110 \rangle \equiv [110], [-1-10], [011], [101], [0-1-1], [-10-1], [1-10], [-101], [-110], [10-1], [01-1], [0-11]$ (12 members)

